

Next-gen 3D genome mapping technologies

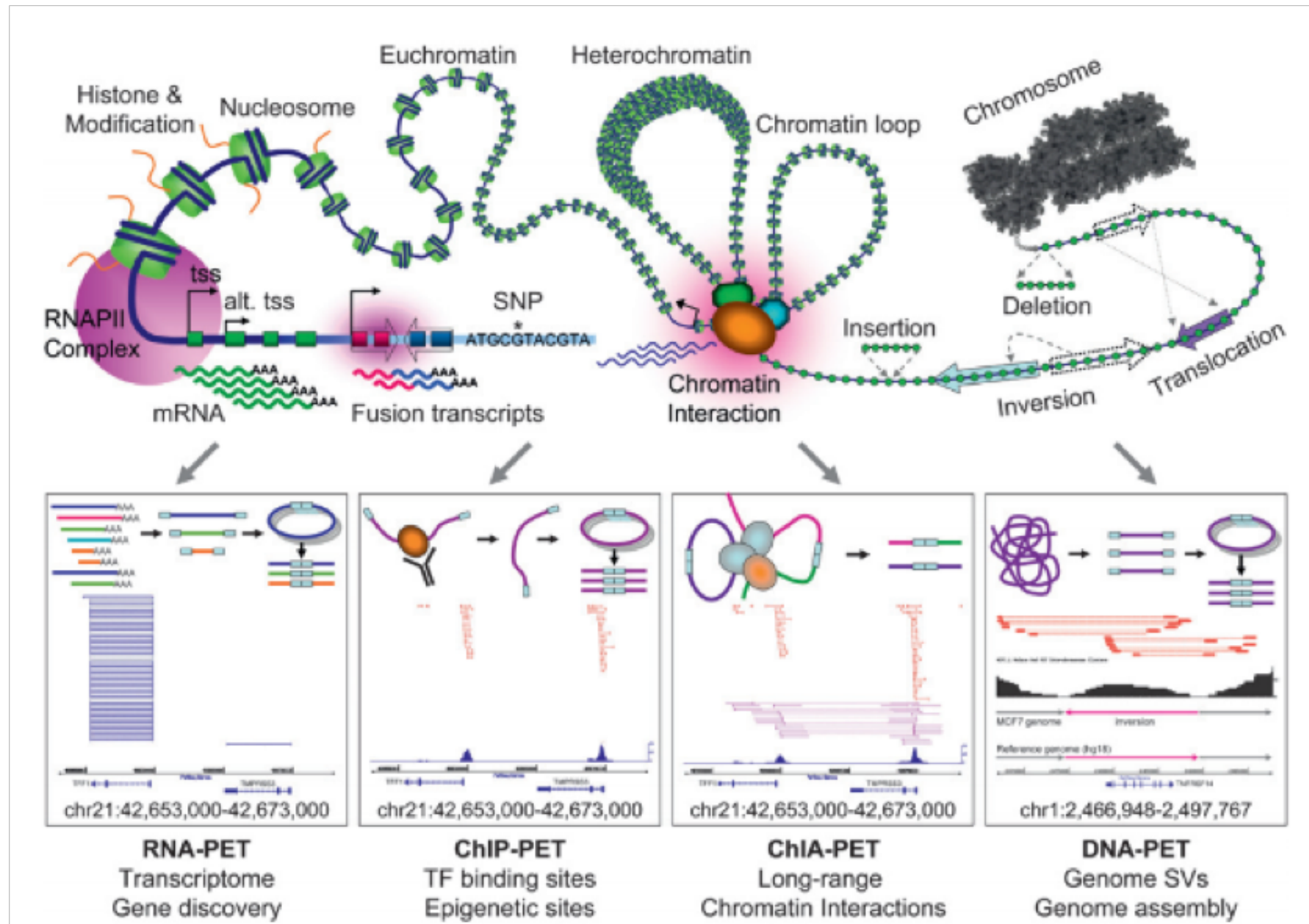
Main content

ChIA-Drop

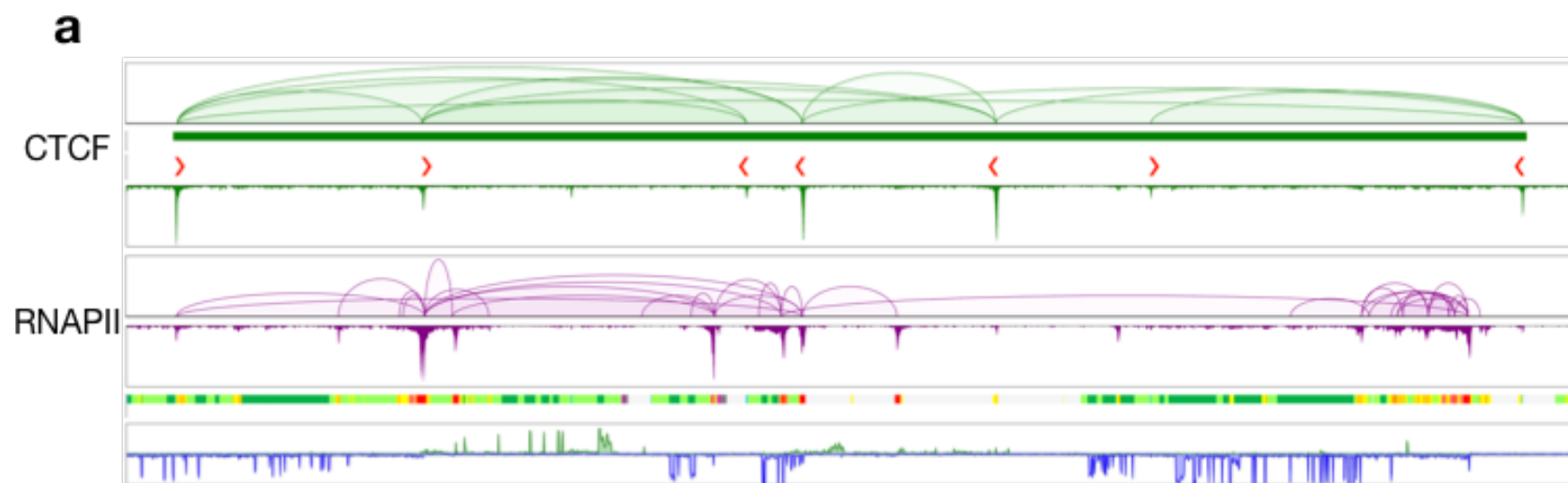
SPRITE

GAM

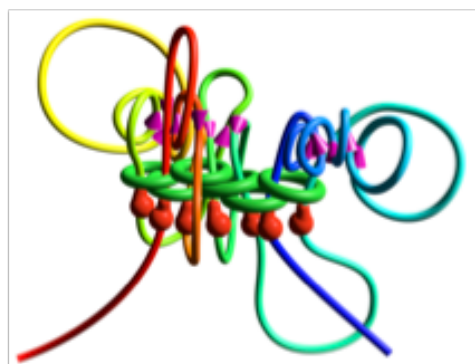
Proximity ligation

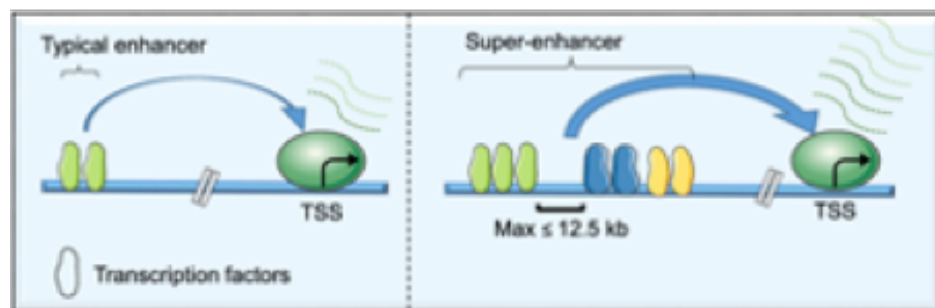
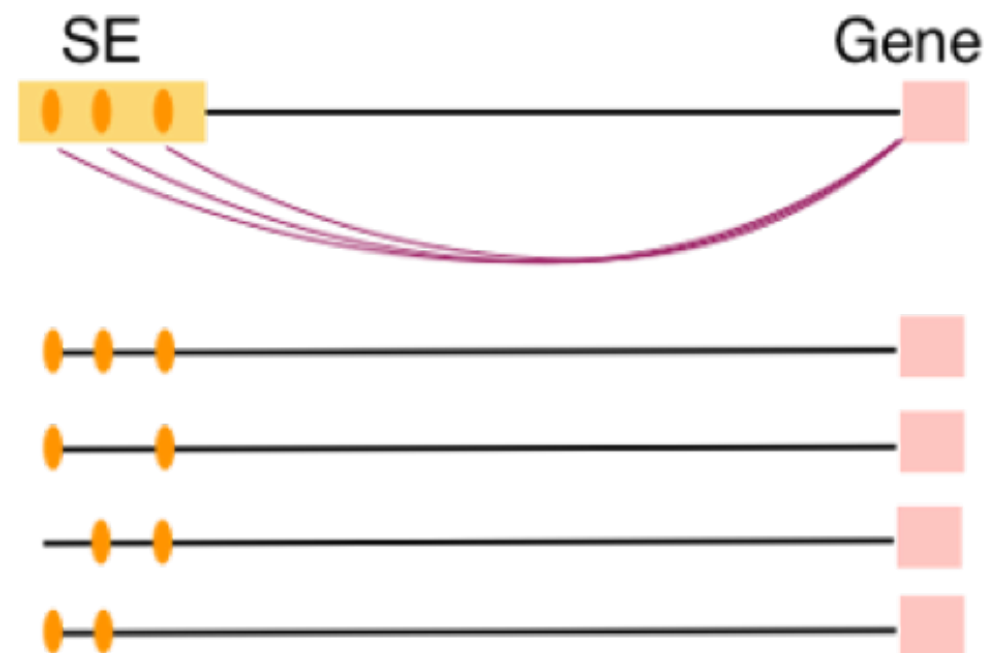
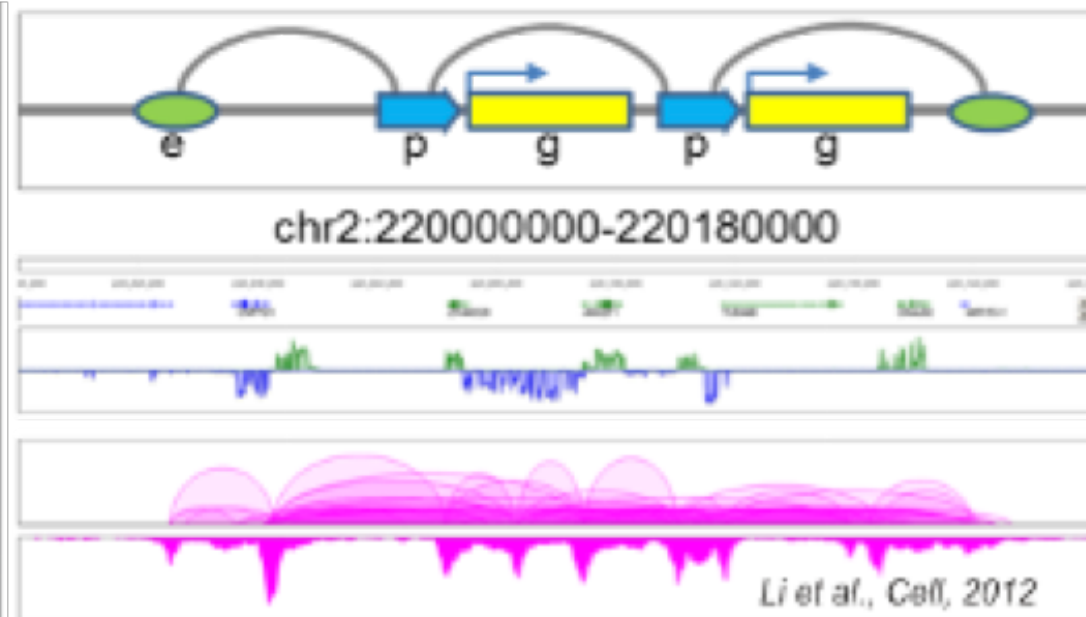
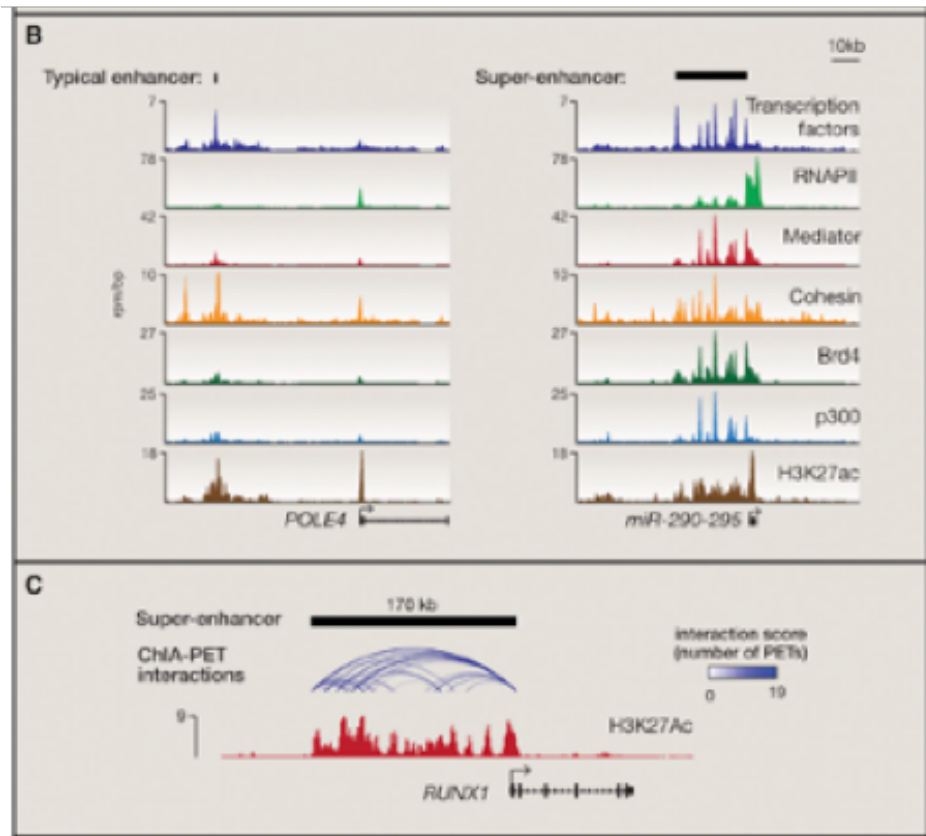


ChIA-PET Data



b

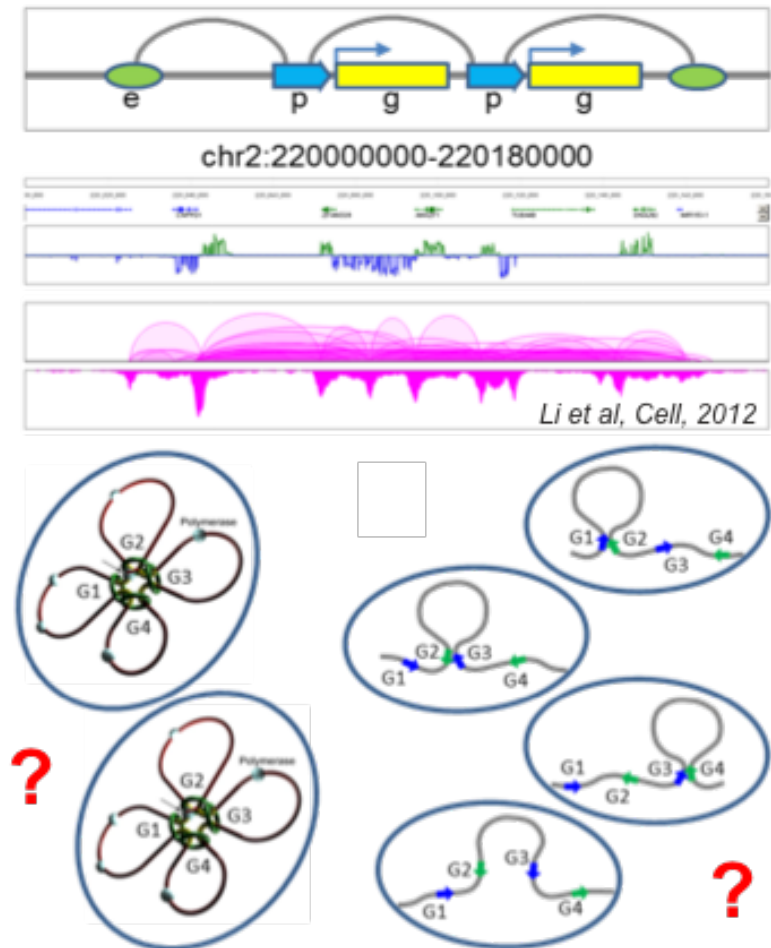




Current limitation in 3D mapping methods

Hi-C and ChIA-PET

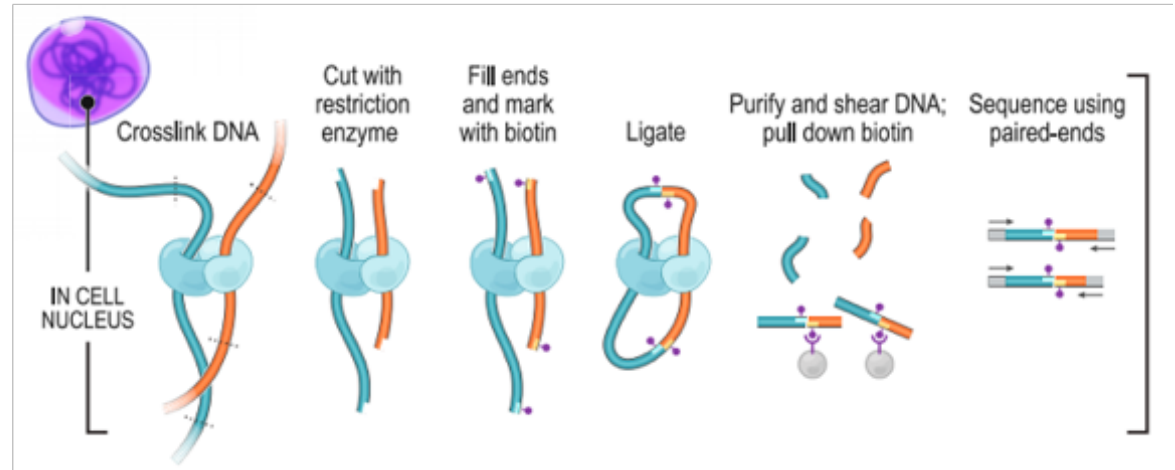
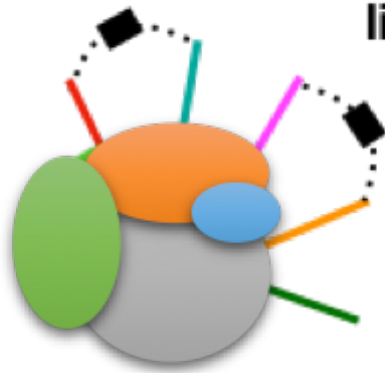
- Dependent on proximity ligation
- Detect only pairwise interactions
- Cannot reveal multiplex interactions



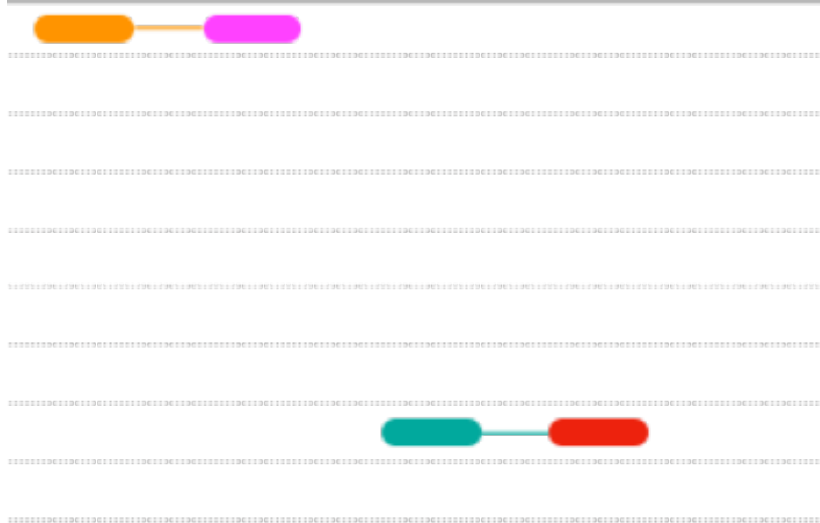
● Goal:

Develop a method that is not dependent on proximity ligation

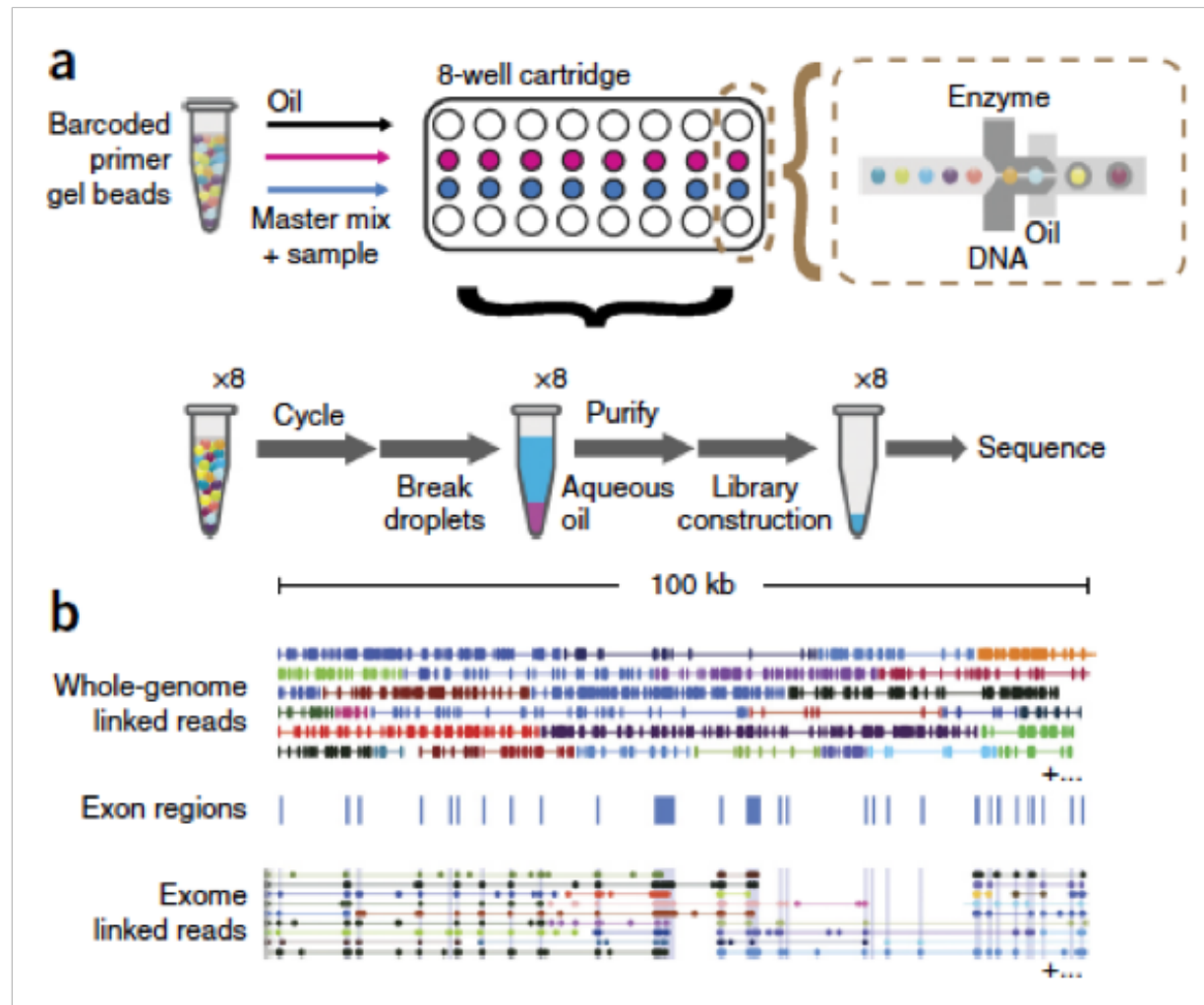
Proximity ligation



Pairwise



Microfluidics system for high molecular weight genomic DNA sequencing



10x Genomics, Chromium system

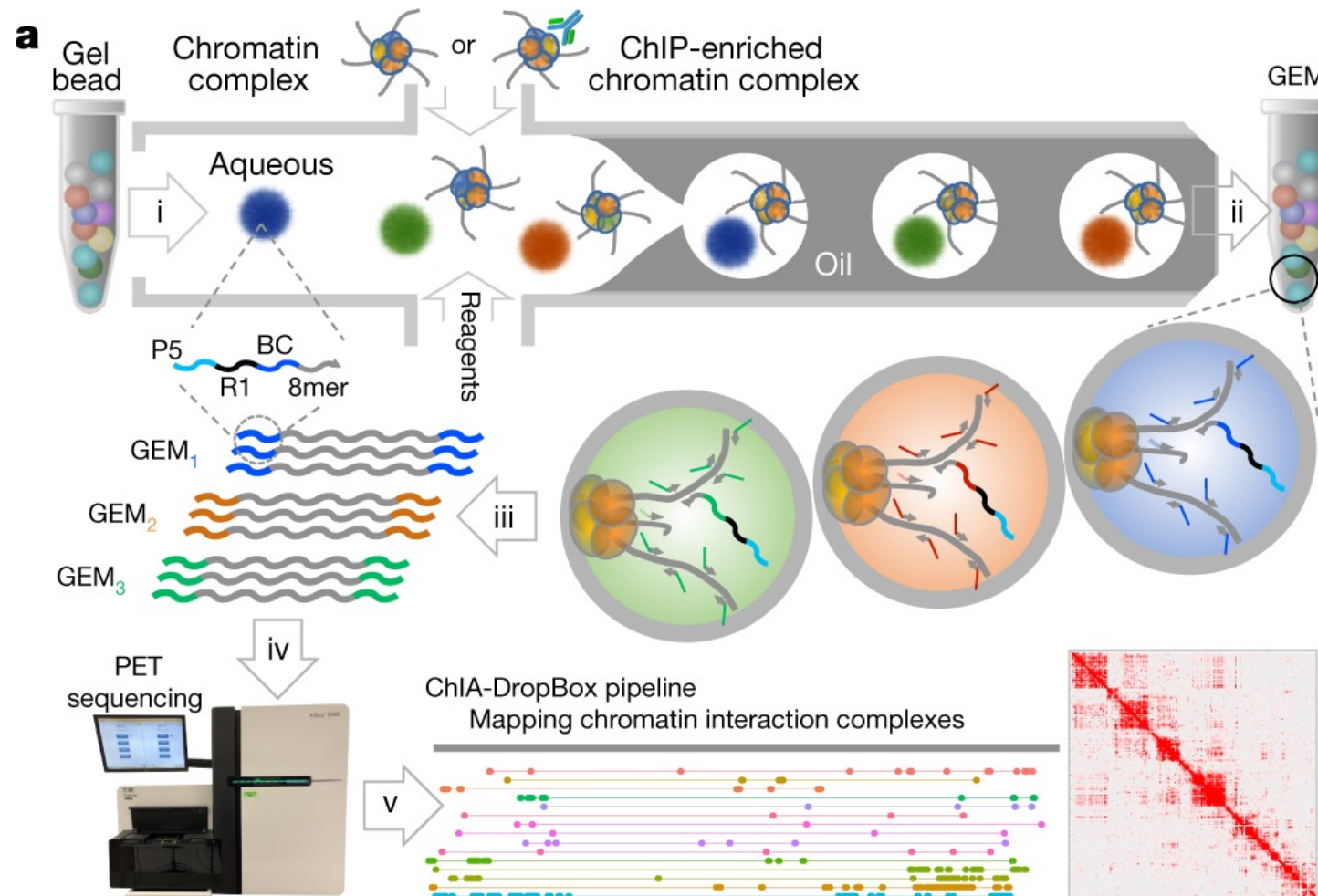
Zheng et al, Nature Biotechnology, 2016

X

What+is+droplet-based+microfluidics .mp4
Dolomite.mp4

Multiplex chromatin interactions with single-molecule precision

Meizhen Zheng¹, Simon Zhongyuan Tian¹, Daniel Capurso¹, Minji Kim¹, Rahul Maurya¹, Byoungkoo Lee¹, Emaly Piecuch^{1,2}, Liang Gong¹, Jacqueline Jufen Zhu^{1,2}, Zhihui Li^{1,3}, Chee Hong Wong¹, Chew Yee Ngan¹, Ping Wang¹, Xiaolan Ruan¹, Chia-Lin Wei¹ & Yijun Ruan^{1,2,4*}



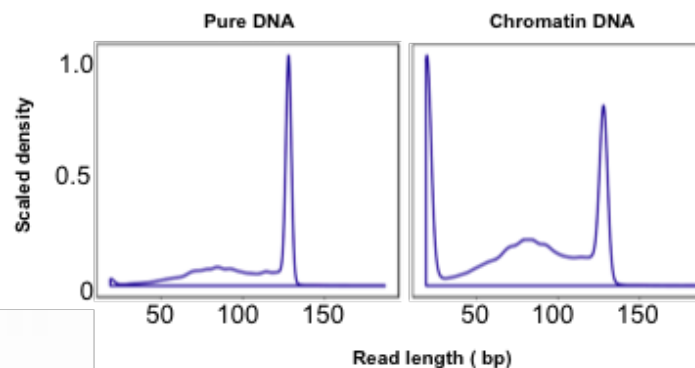
Characterization of Chromium microfluidics system for chromatin DNA sequencing

Question?

Could the microfluidics system be applied for chromatin DNA barcoding and amplification?

Pure DNA vs Chromatin DNA

	Pure DNA	Chromatin DNA
Total reads	2,545,248	2,191,091
GEM barcode with reads	1,064,328	686,493
Reads with GEM barcode	2,355,569	1,970,016
Uniquely mapped reads	2,261,170	1,155,343
Uniquely mapped reads (%)	95.99%	58.65%



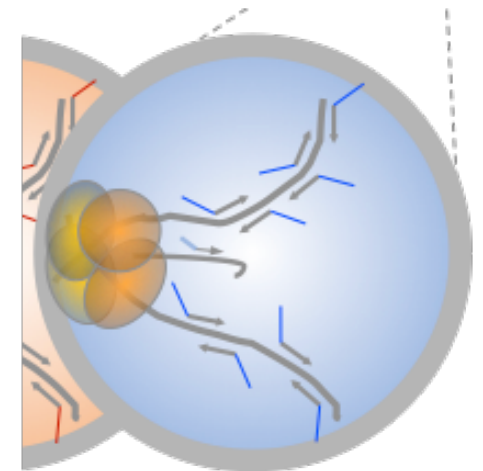
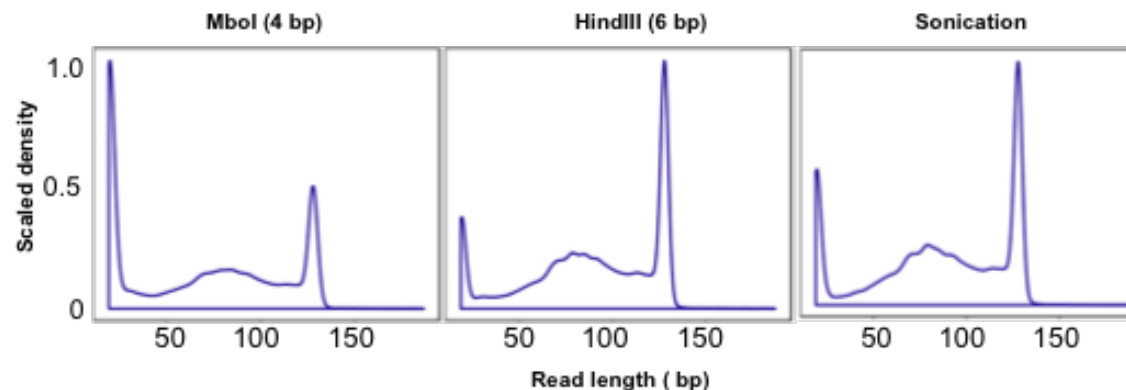
Characterization of Chromium microfluidics system for chromatin DNA sequencing

Question?

What is the optimal chromatin DNA fragment length for this sequencing technology?

Different chromatin fragment length

	4 bp cutter (Mbol)	6 bp cutter (HindIII)	Sonication
Total reads	4,656,784	4,868,248	2,274,570
GEM barcode with reads	845,883	1,044,653	748,048
Reads with GEM barcode	4,216,577	4,460,036	2,043,330
Uniquely mapped reads	1,782,446	3,178,376	1,517,719
Uniquely mapped reads (%)	42.27%	71.26%	74.28%



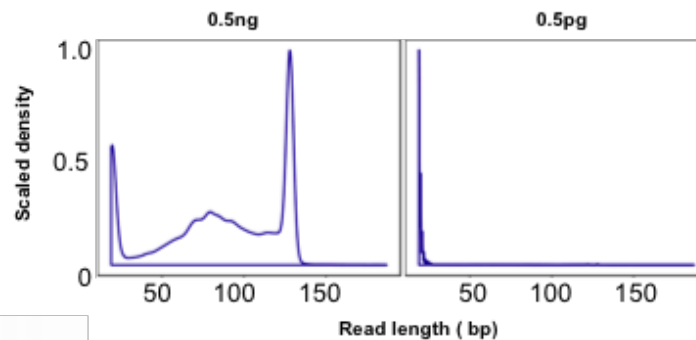
Characterization of Chromium microfluidics system for chromatin DNA sequencing

Question?

What is the optimal loading quantity of chromatin DNA for this sequencing technology?

Different loading quantity of chromatin DNA

	0.5 ng	0.5 pg
Total reads	2,274,570	1,780,274
GEM barcode with reads	748,048	519,010
Reads with GEM barcode	2,043,330	1,601,257
Uniquely mapped reads	1,517,719	456,471
Uniquely mapped reads (%)	74.28%	28.51%



Characterization of Chromium microfluidics system for chromatin DNA sequencing

Question?

Can we estimate the percentage of droplets that contain more than one chromatin complex? (false positive interactions)

Inter-species chromatin samples

<i>Drosophila</i> : Human	Cell # (1:1)	Chromatin (1:1)
Total reads	1,992,569	4,656,784
GEM barcode with reads	669,843	845,883
Reads with GEM barcode	1,809,743	4,216,577
Uniquely mapped reads	808,269	1,782,446
GEM with fragments*	192,377	279,127
<i>Drosophila</i>	9,149	142,975
(%)	4.76%	51.22%
Human	181,956	121,920
(%)	94.58%	43.68%
Inter-species	1,272	14,232
(%)	0.66%	5.10%

Pure DNA vs Chromatin DNA

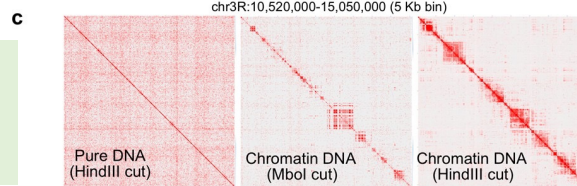
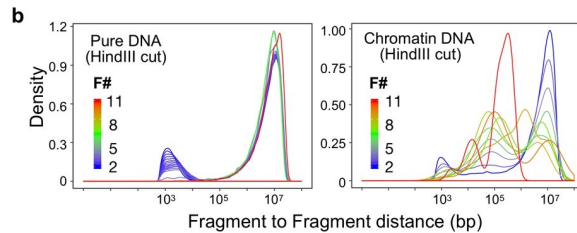
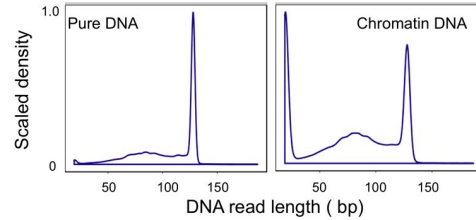
Chromatin DNA length?

Loading amount?

System control?

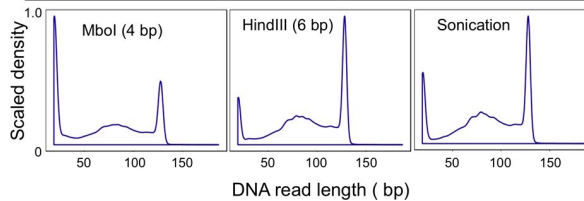
a Pure DNA vs Chromatin DNA

	Pure DNA	Chromatin DNA
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d Different chromatin fragment length

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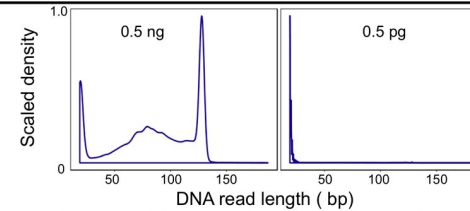


e Sequencing data of Mbol vs HindIII cuts

	4 bp cutter (Mbol)	6 bp cutter (HindIII)
Total reads	98,399,745	105,814,667
Reads with GEM barcode	87,779,074	95,271,125
GEM barcode	1,919,641	1,914,282
Uniquely mappable reads	39,246,891	65,051,724
Uniquely mappable reads (%)	44.70%	68.30%
F = 1	616,750	1,837,201
F = 2	40,401	406,107
F = 3	4,594	100,538
F = 4	1,112	29,781
F = 5	466	11,347
F = 6	287	5,350
F = 7	164	3,089
F = 8	104	1,944
F = 9	77	1,312
F = 10	73	944
F = 11	51	728
F = 12	41	592
F = 13	33	474
F = 14	23	399
F = 15	26	329
F > 15	151	2,522

f Different loading quantity of chromatin DNA

	0.5 ng	0.5 pg
Total reads	2,274,570	1,780,274
GEM barcode with reads	748,048	519,010
Reads with GEM barcode	2,043,330	1,601,257
Uniquely mapped reads	1,517,719	456,471
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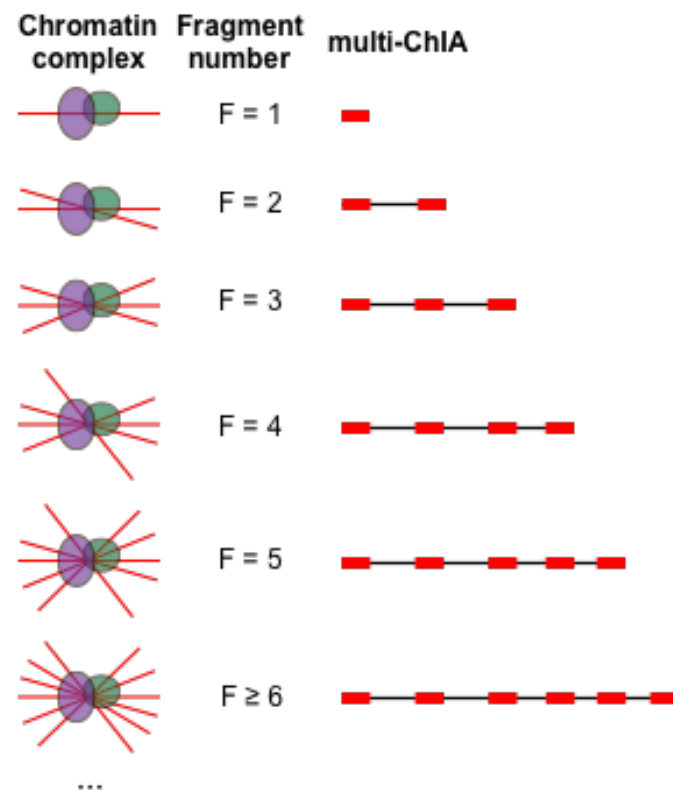
g Sequencing data of Inter-species chromatin samples

	Cell # (1:1)	Chromatin (1:1)
Drosophila : Human		
Total reads	1,992,569	4,656,784
GEM barcode with reads	669,843	845,883
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GEM with fragments	192,377	279,127
Drosophila		
(%)	9.149	142,975
Human		
(%)	4.76%	51.22%
(%)	181,956	121,920
(%)	94.58%	43.68%
Inter-species		
(%)	1,272	14,232
(%)	0.66%	5.10%

Multi-ChIA data from *Drosophila* S2 cells

- S2 cell is a good model system for epigenome and 3D genome
- Small genome size

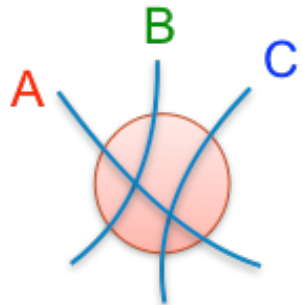
a



b

Complex class	Complex #
F = 2	1,549,918
F = 3	789,247
F = 4	387,083
F = 5	190,036
F = 6	96,229
F = 7	50,146
F = 8	27,318
F = 9	15,817
F = 10	9,547
F = 11	6,052
F = 12	4,151
F = 13	3,006
F = 14	2,123
F = 15 - 19	5,833
F = 20 - 29	3,960
F = 30 - 39	1,513
F = 40 - 49	742
F = 50 - 99	1,059
F = 100 - 199	265
F = 200 - 499	41
F = 500 - 999	1
In total	3,144,087

Validating multi-ChIA data with Hi-C data

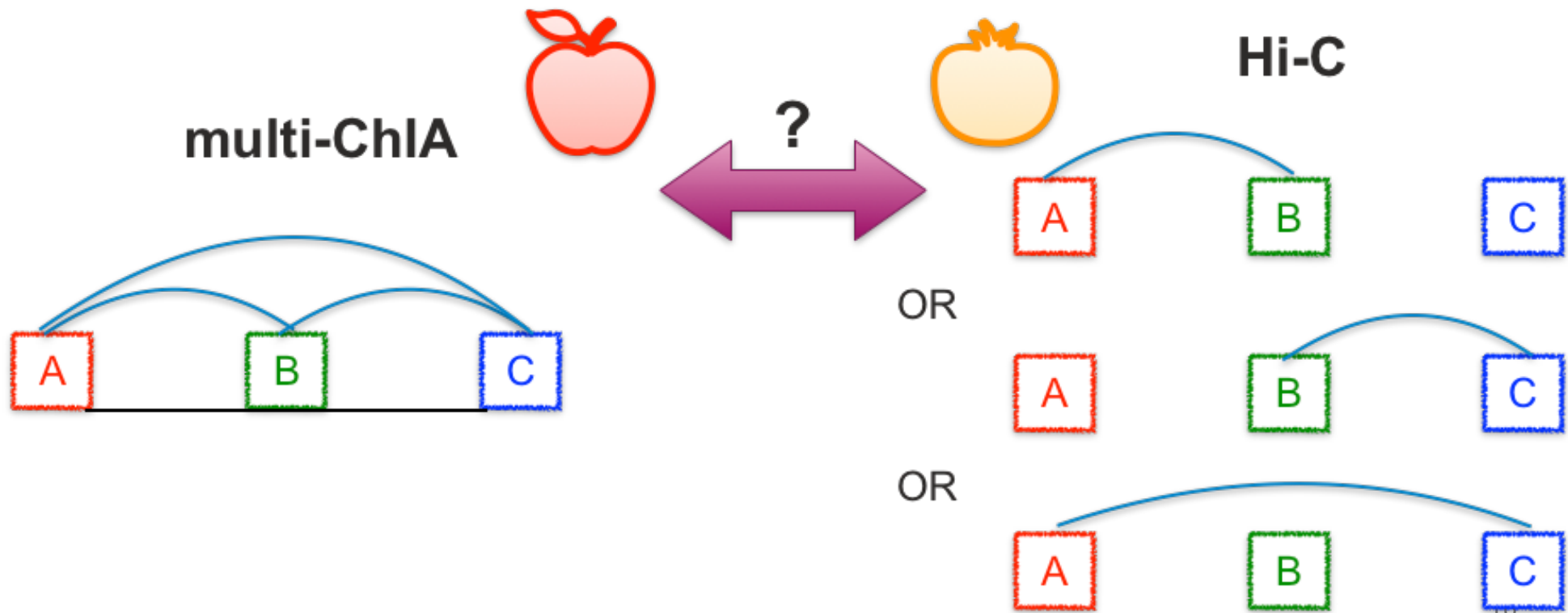


Chromatin Complex

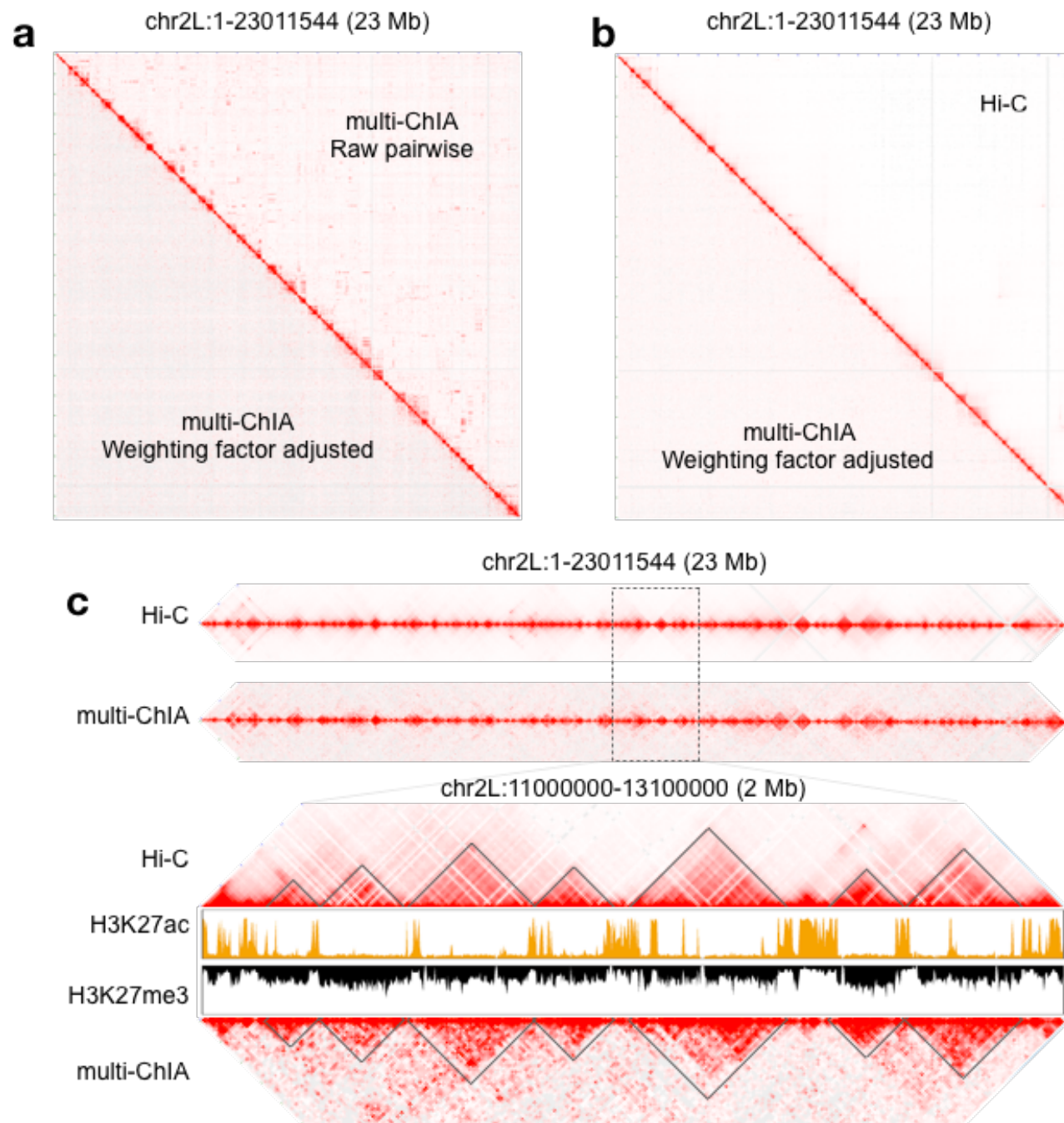
Problem: A complex with n fragments has $n(n-1)/2$ possible pairs

Example: $n=3$, 3 pairs; $n=10$, 45 pairs

Solution: simulate proximity ligation events by
weighting factor or **random sampling**

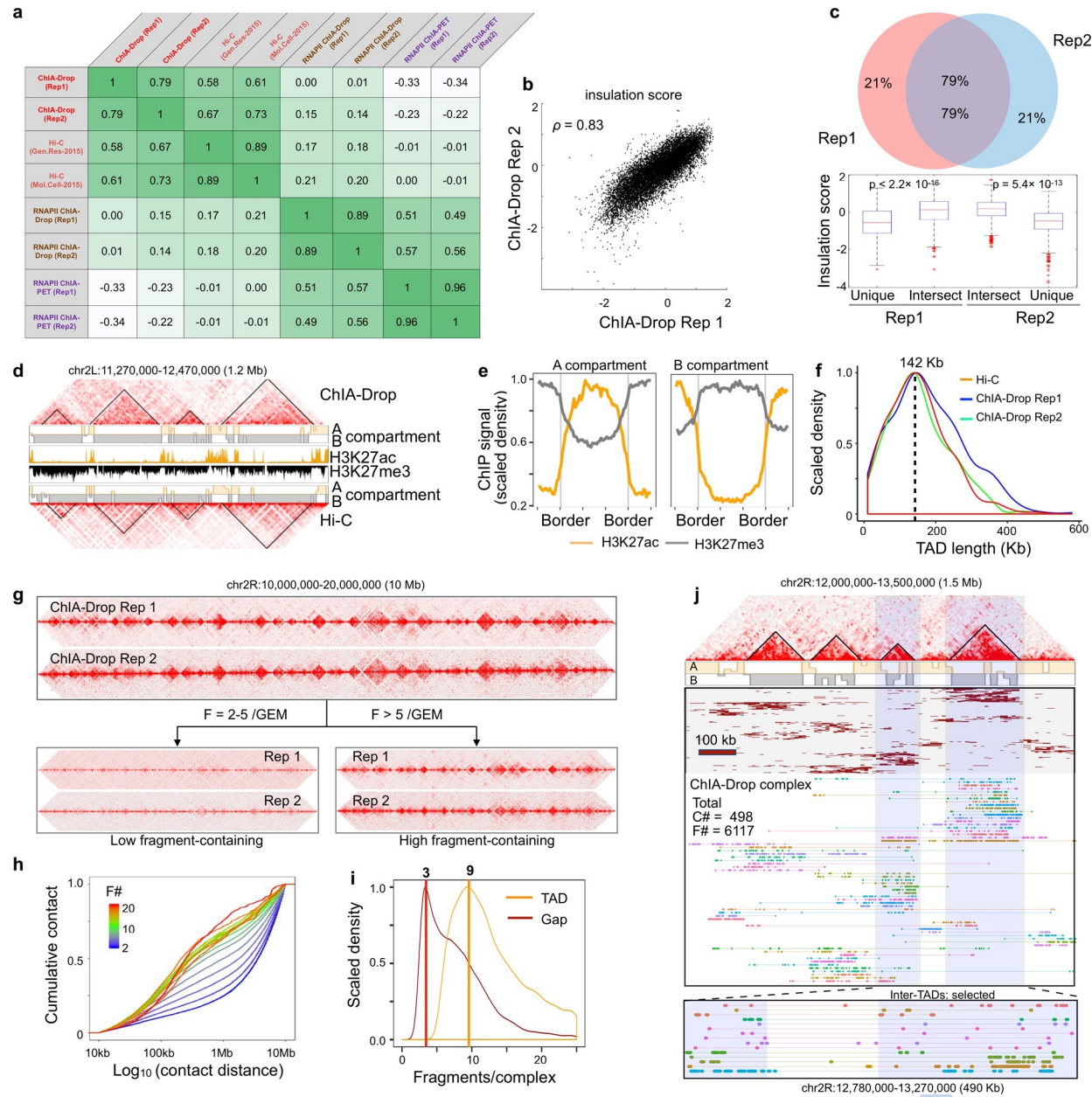


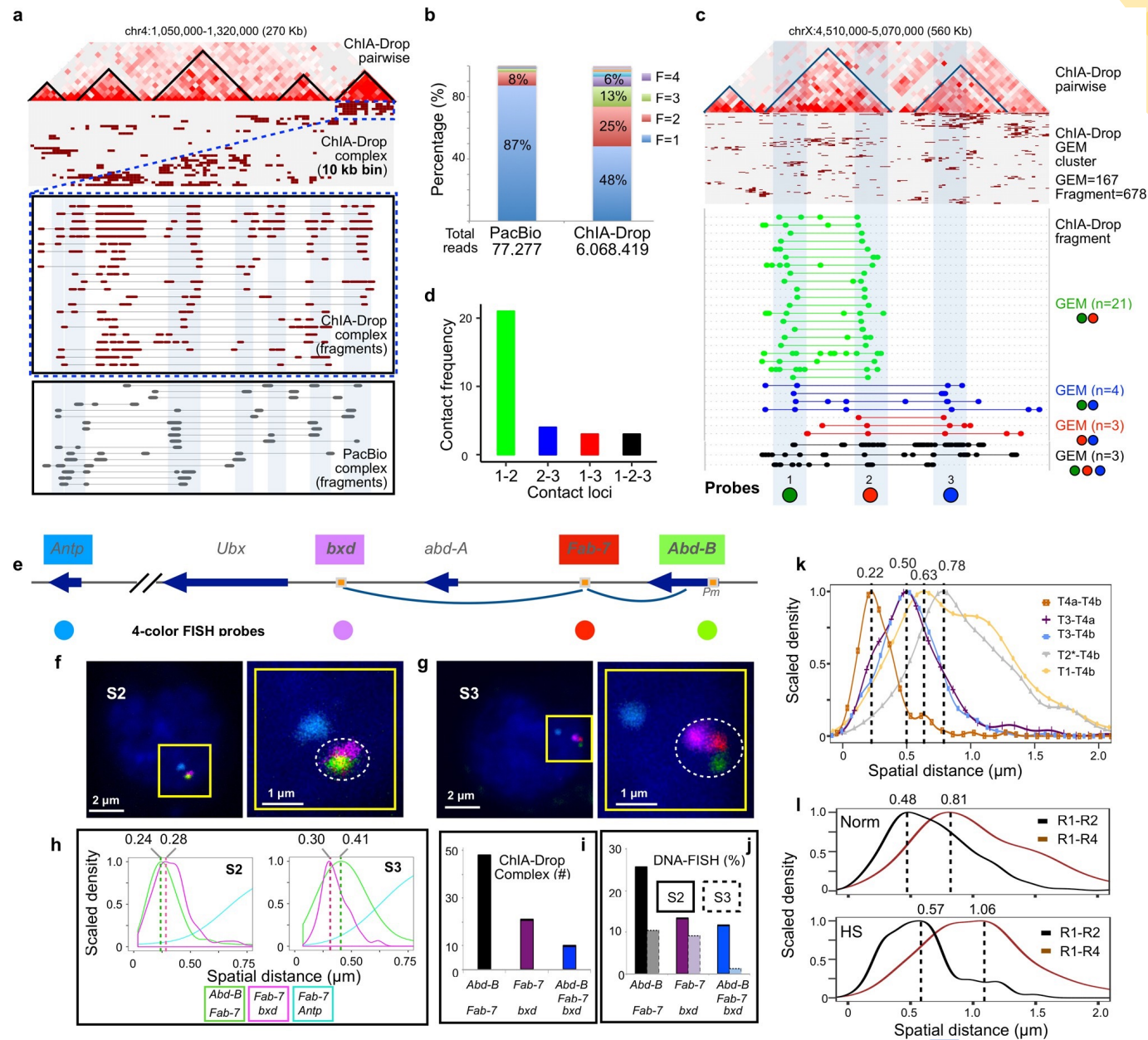
Comparison of multi-ChIA pairwise data with Hi-C data



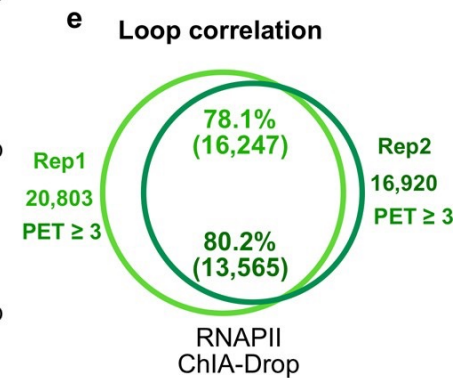
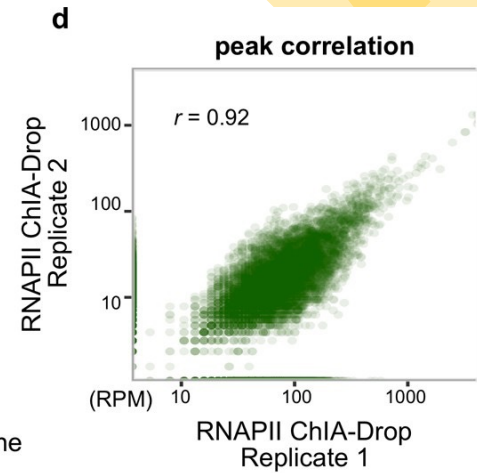
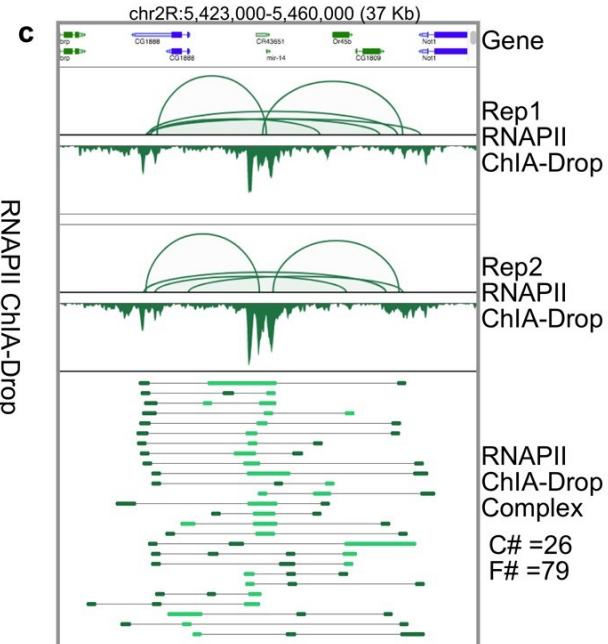
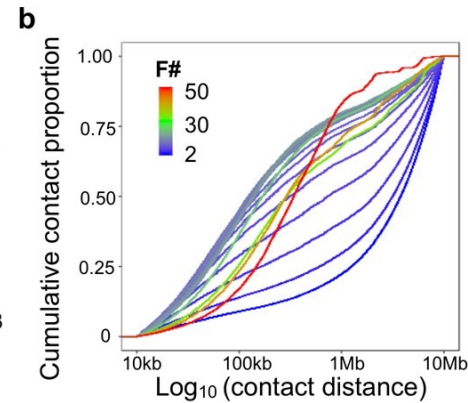
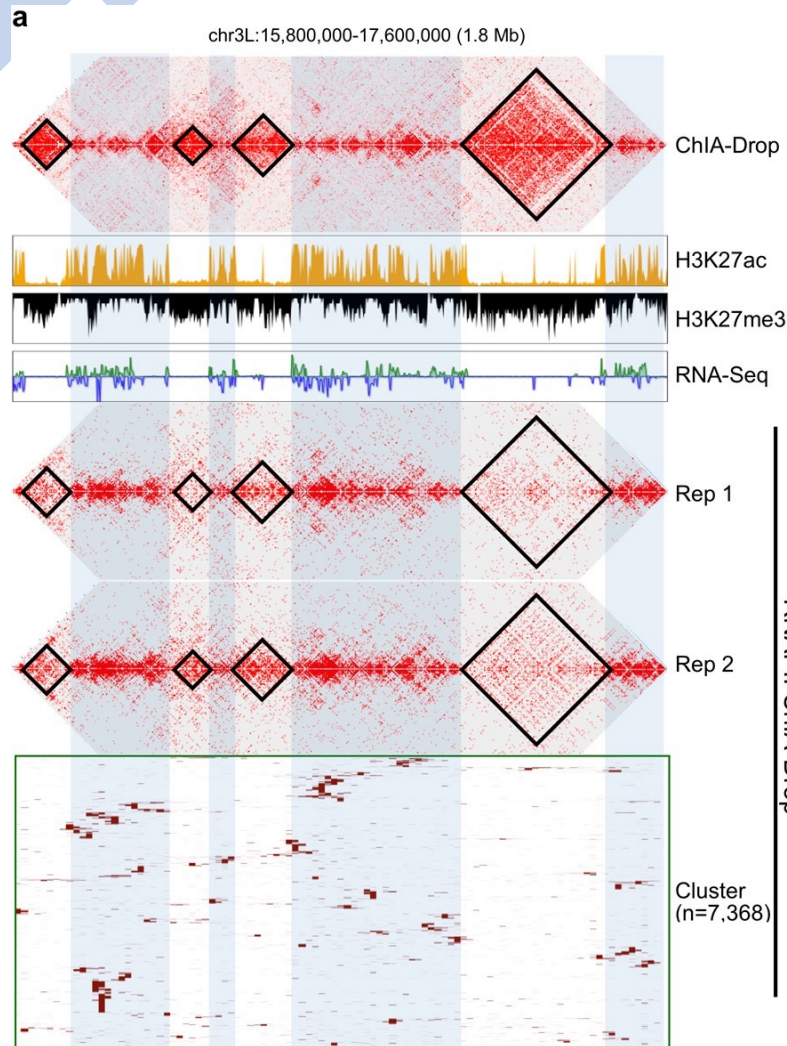
Biological replicates

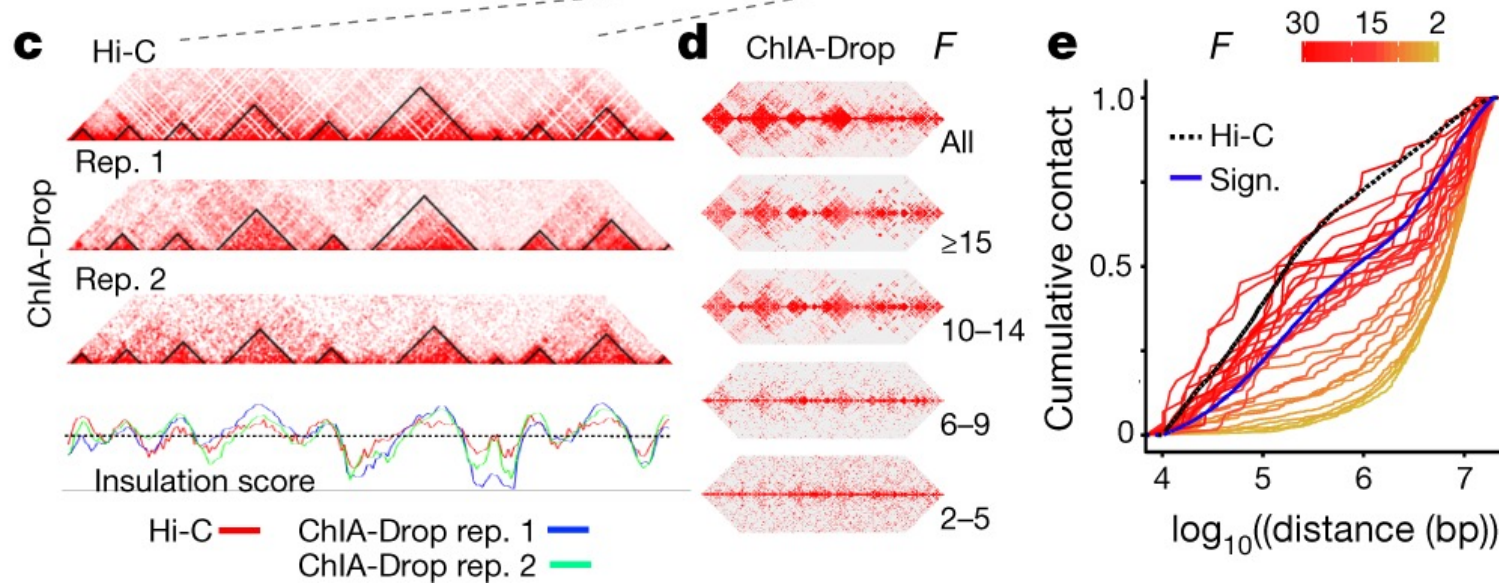
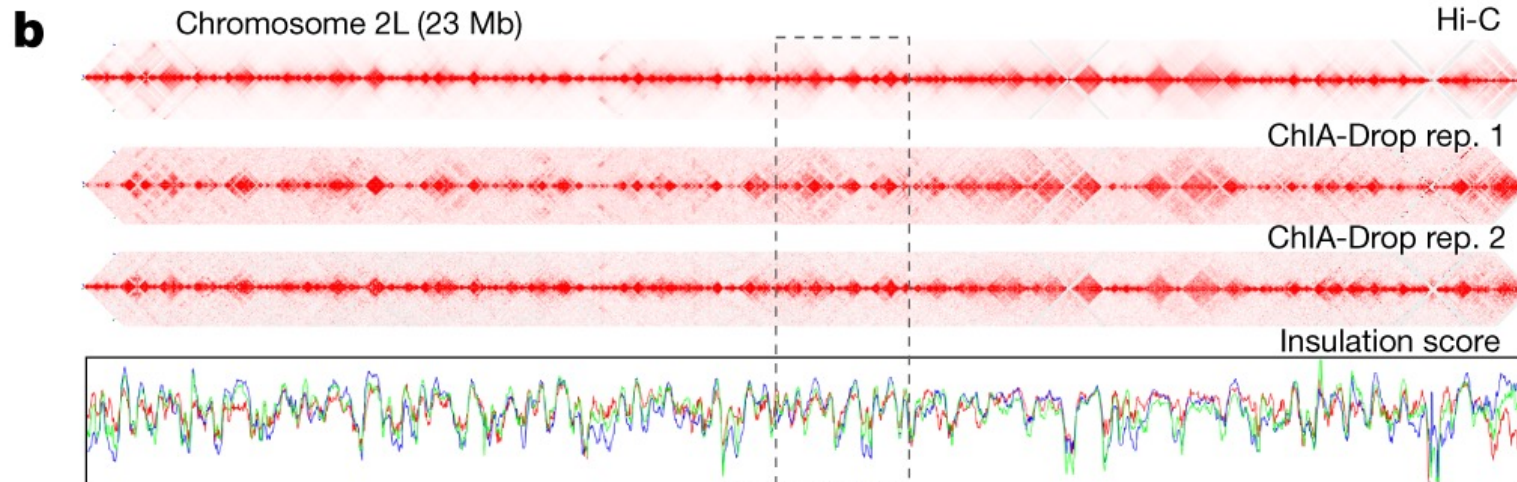
ChIA-Drop vs Hi-C

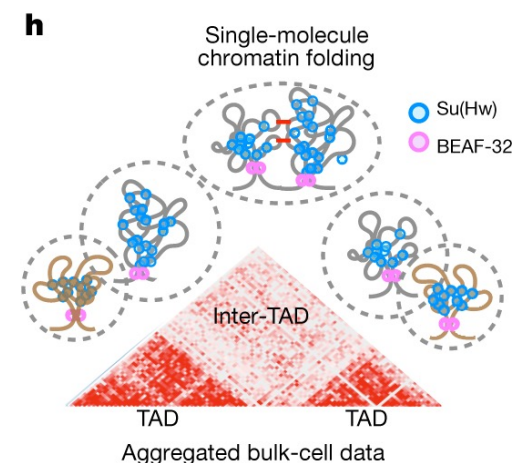
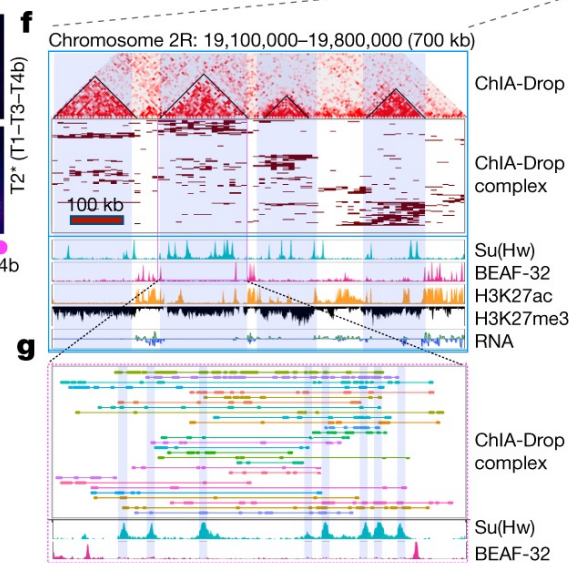
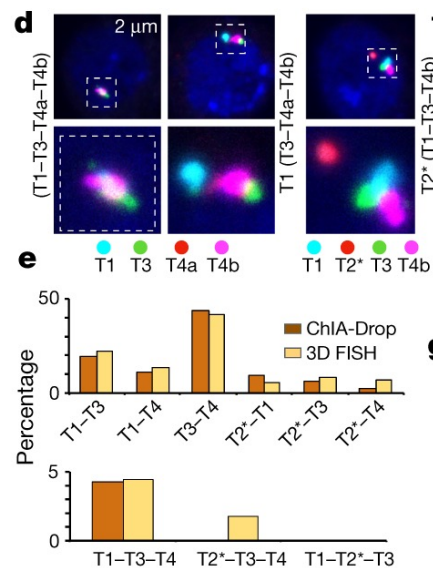
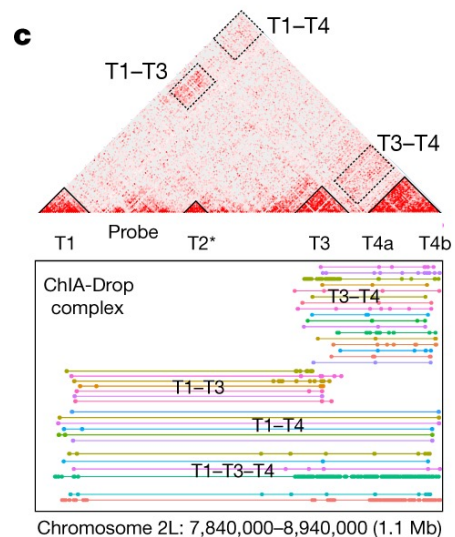
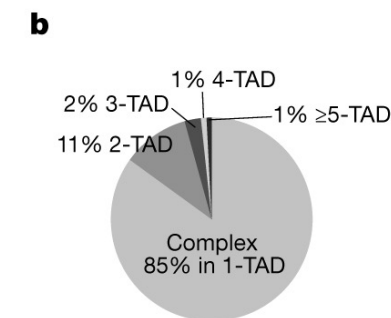
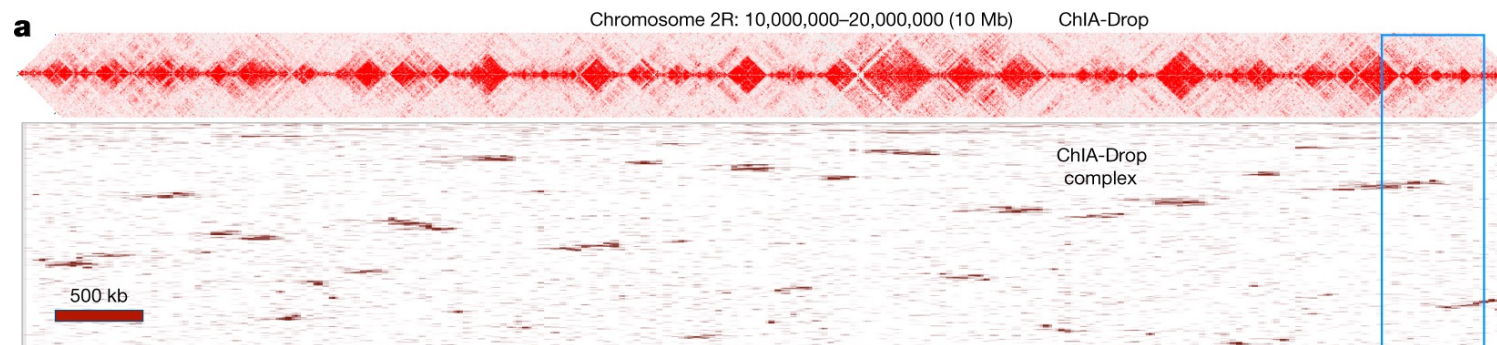
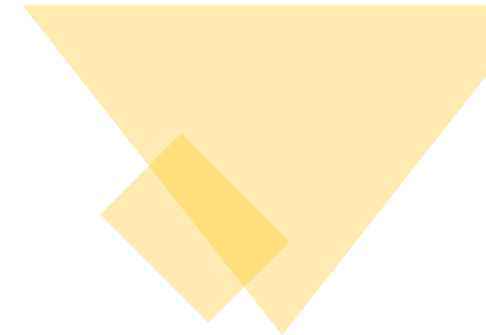
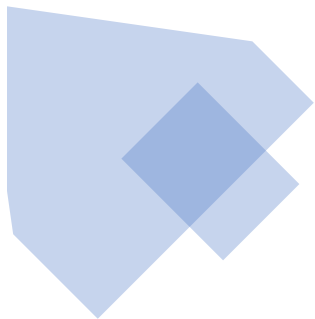


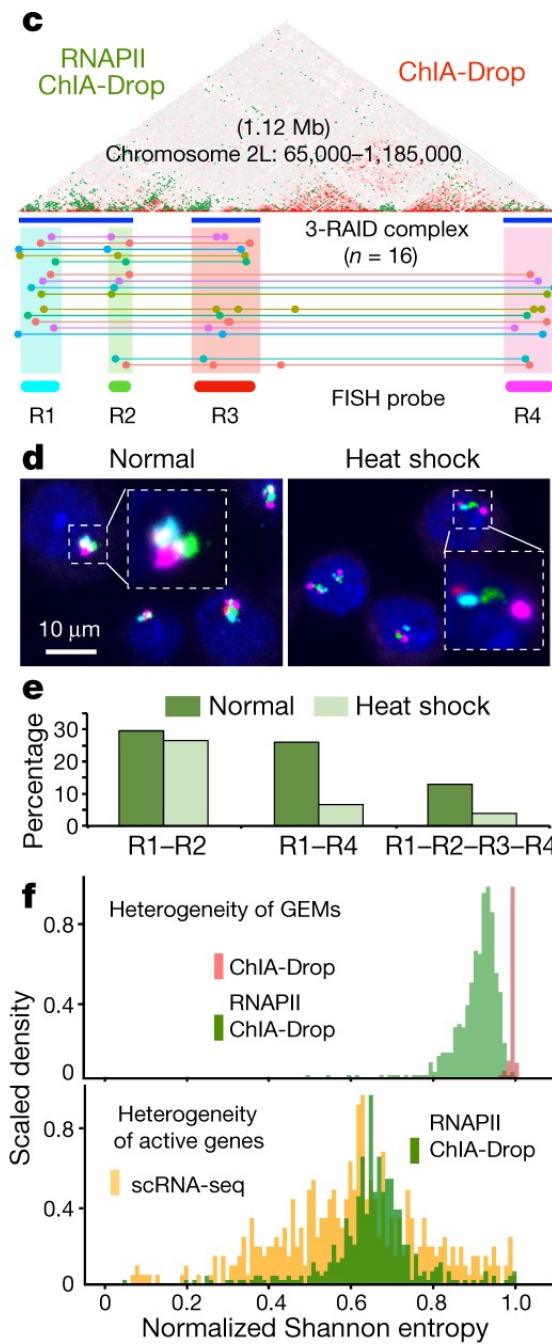
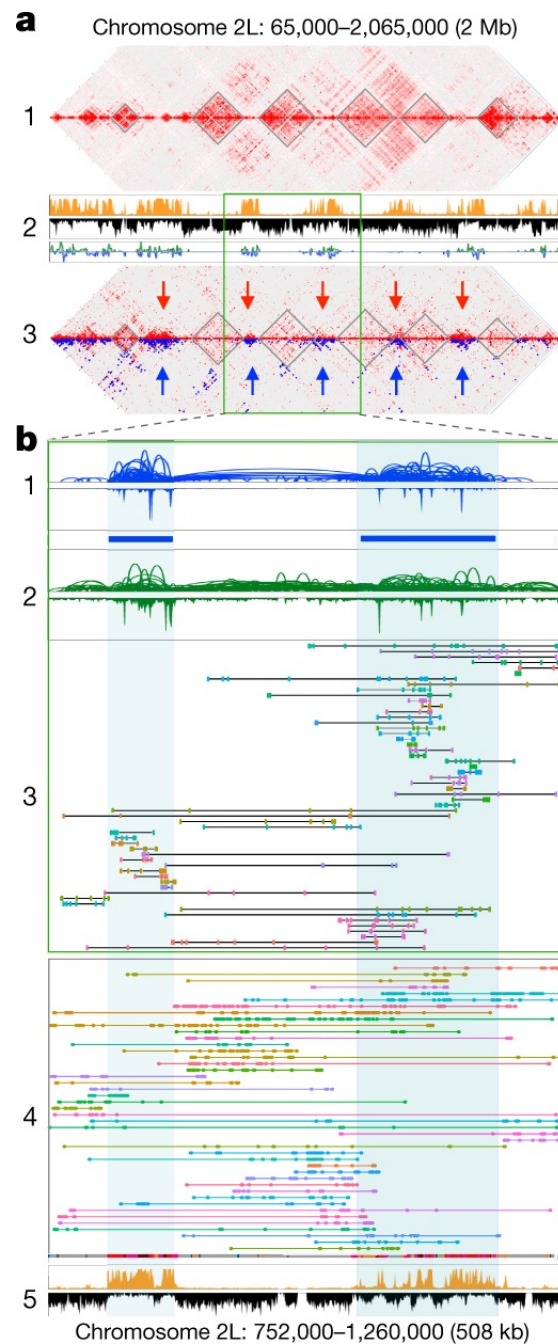


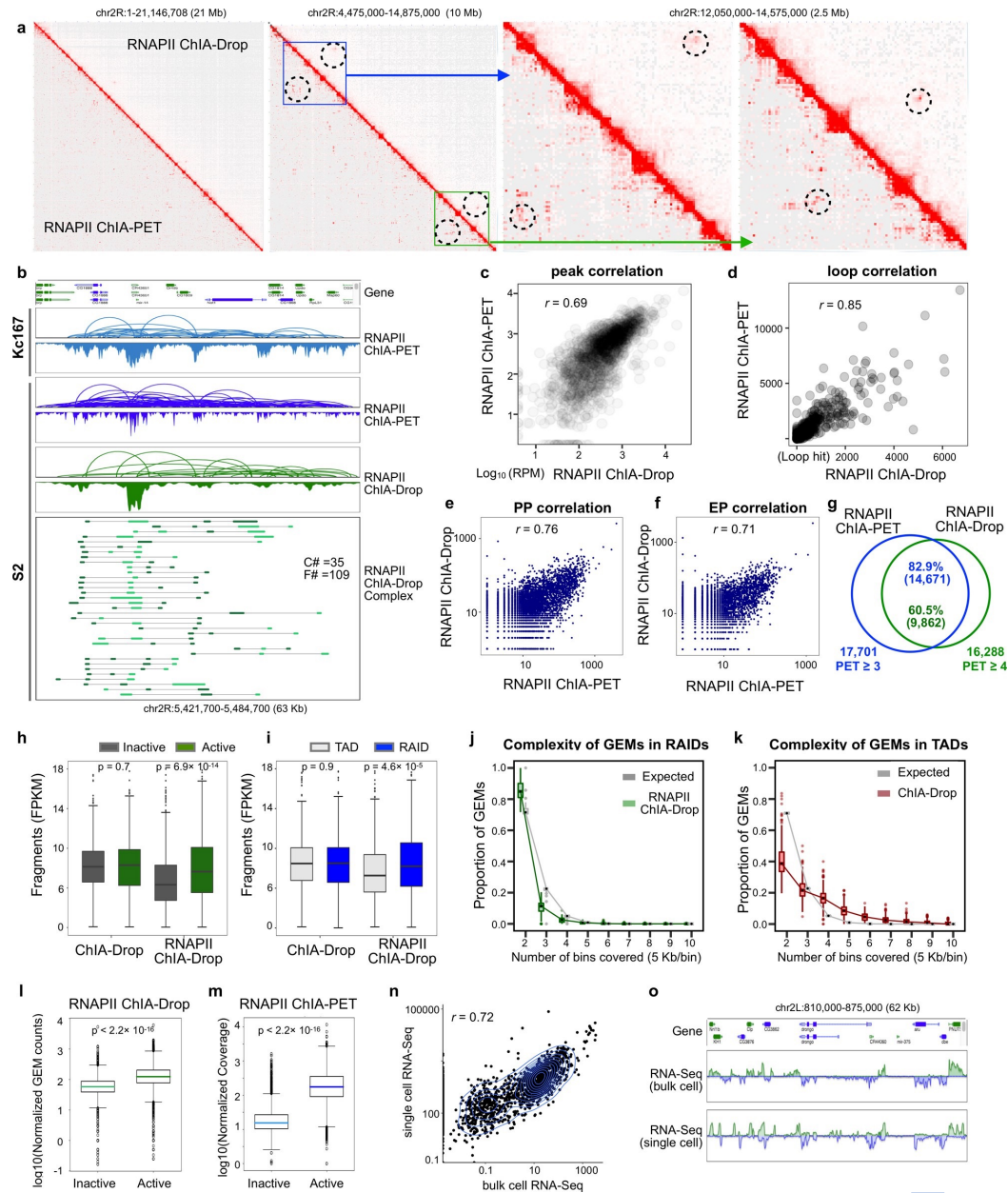
ChIA-Drop VS RNAPII ChIA-Drop

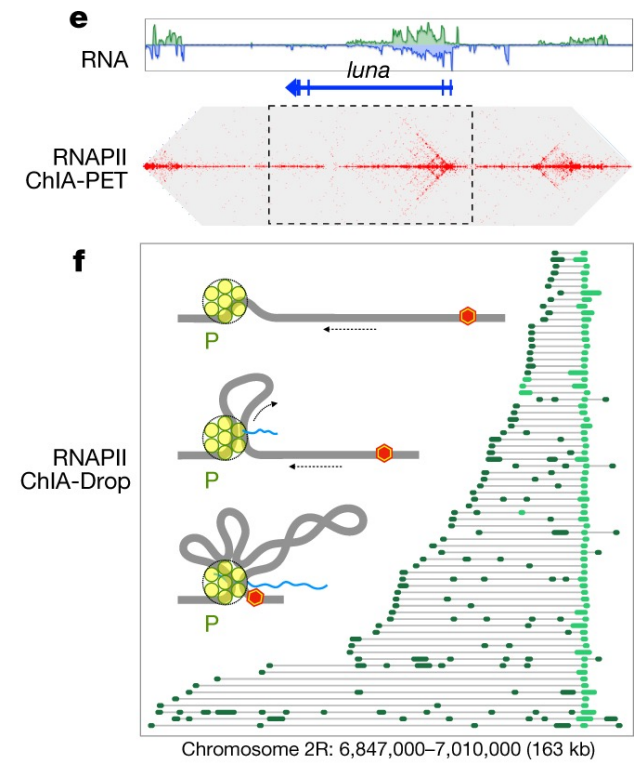
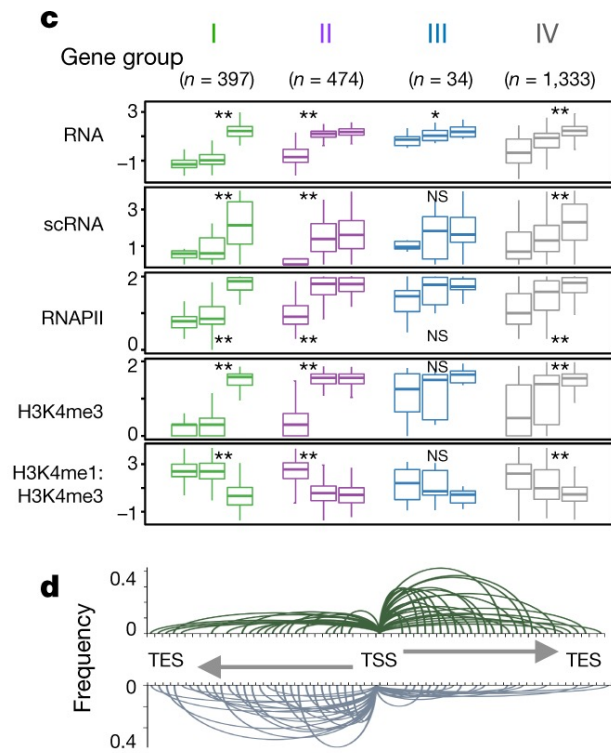
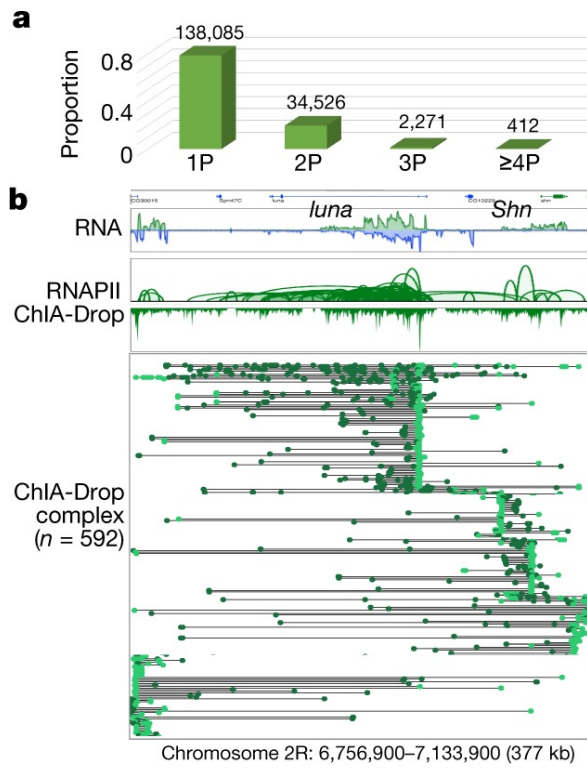


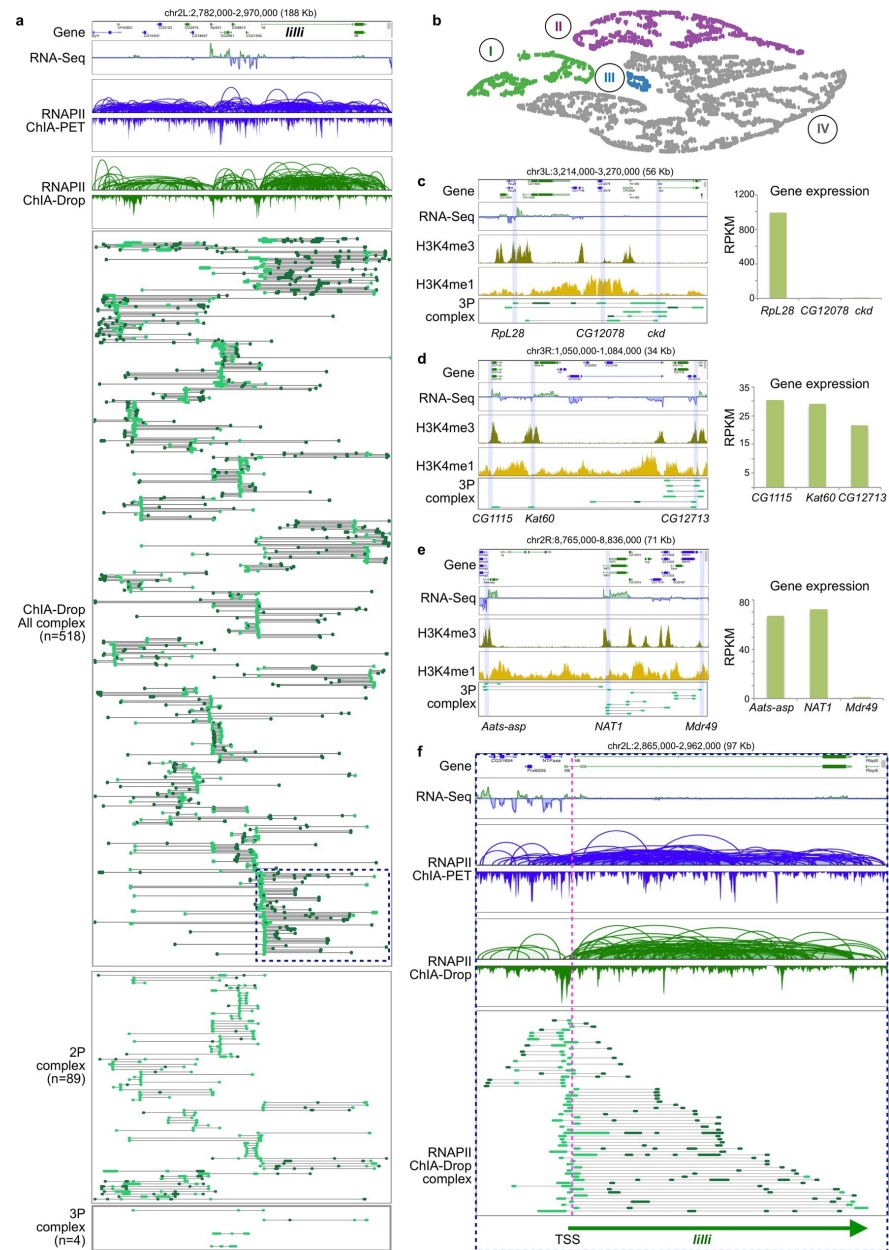


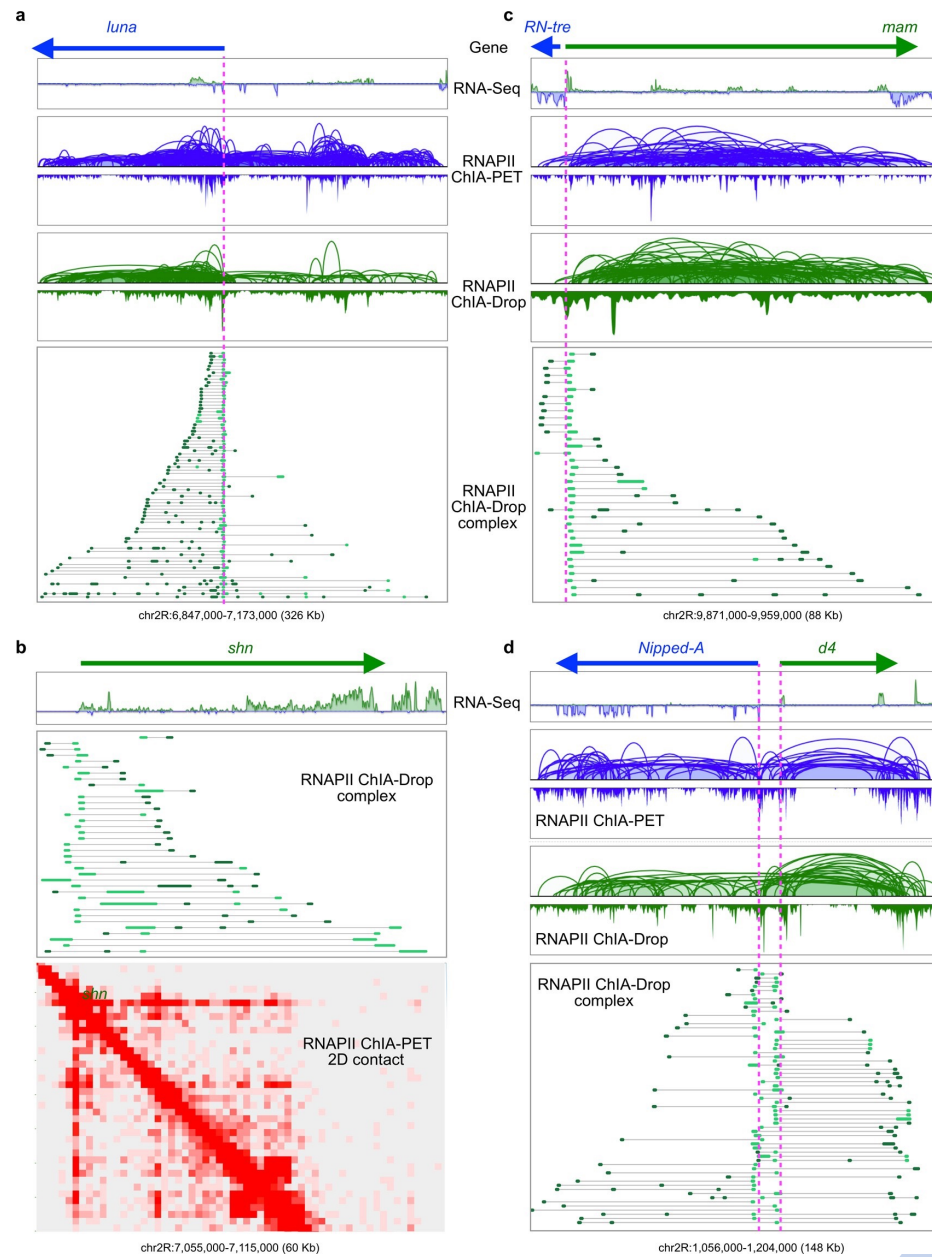






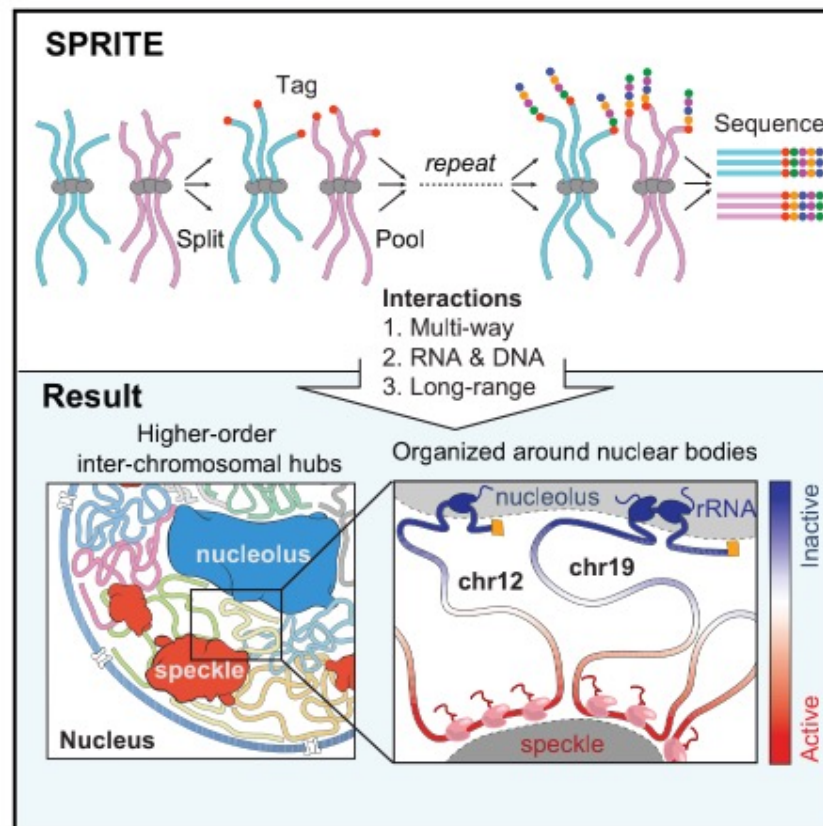






Higher-Order Inter-chromosomal Hubs Shape 3D Genome Organization in the Nucleus

Graphical Abstract



Authors

Sofia A. Quinodoz, Noah Ollikainen, Barbara Tabak, ..., Patrick McDonel, Manuel Garber, Mitchell Guttman

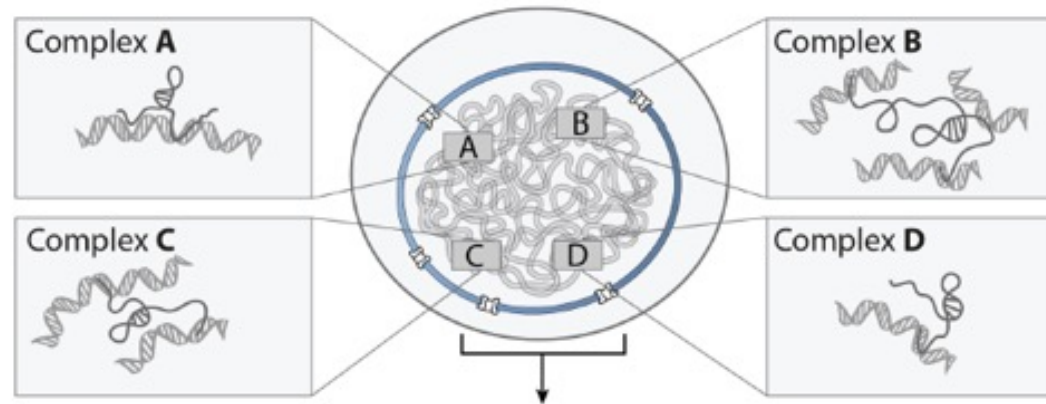
Correspondence

mguttman@caltech.edu

In Brief

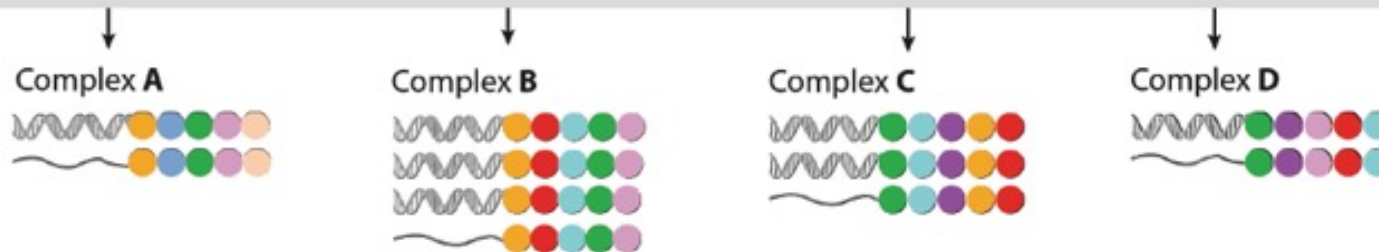
SPRITE enables genome-wide detection of multiple simultaneously occurring higher-order DNA interactions within the nucleus and provides a global picture of inter-chromosomal spatial arrangement around nuclear bodies.

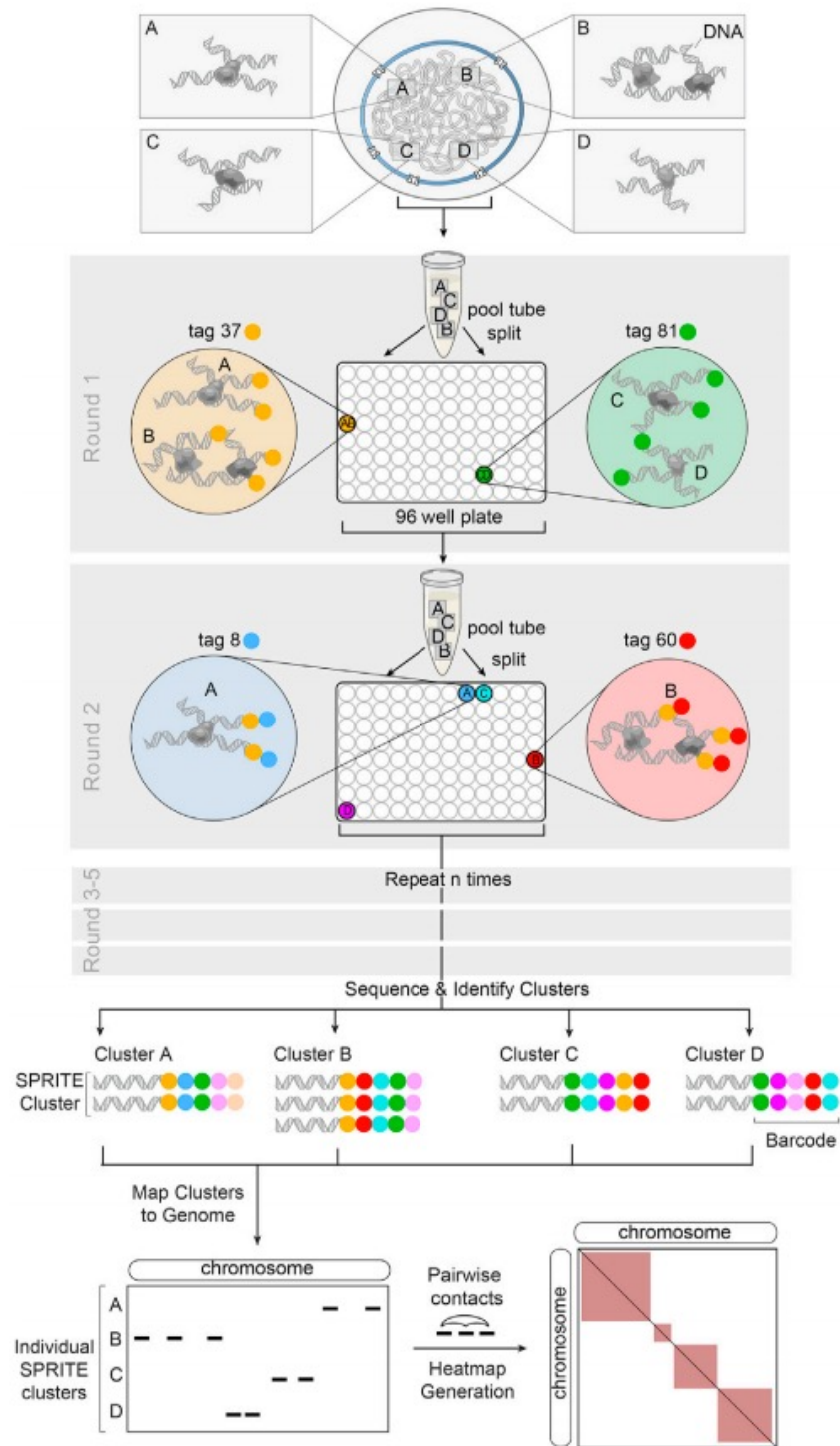
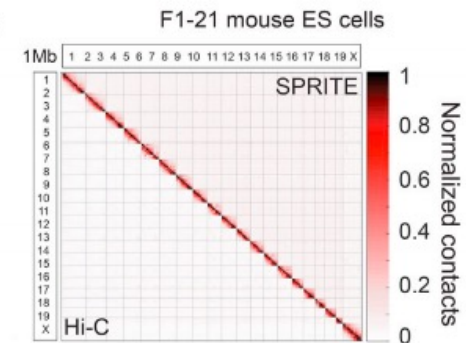
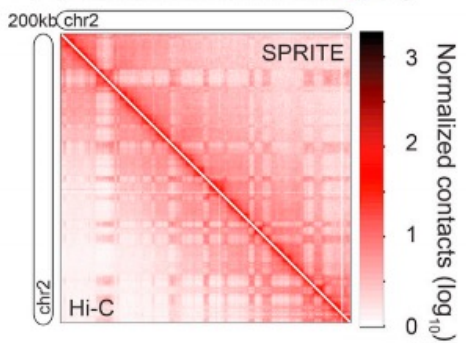
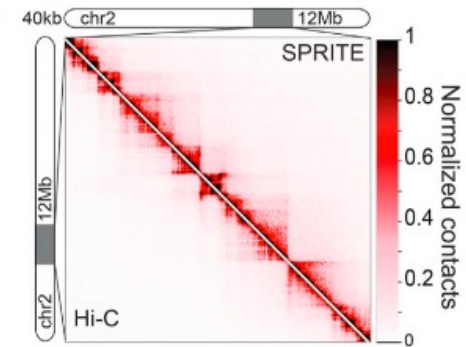
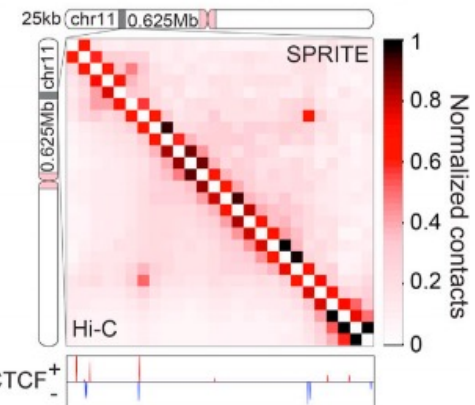
Interacting RNAs & DNAs can be tagged with split-pool barcoding

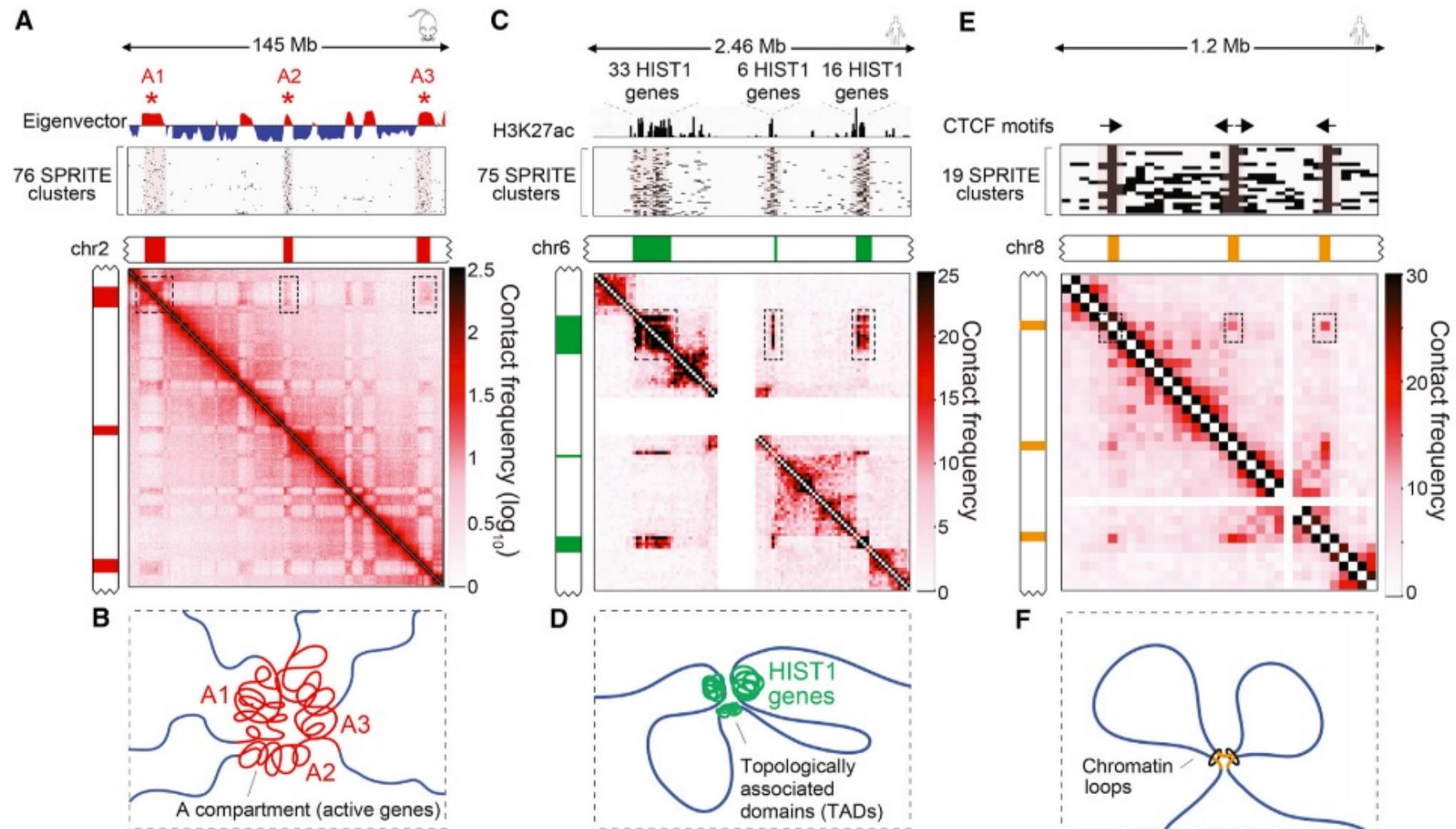


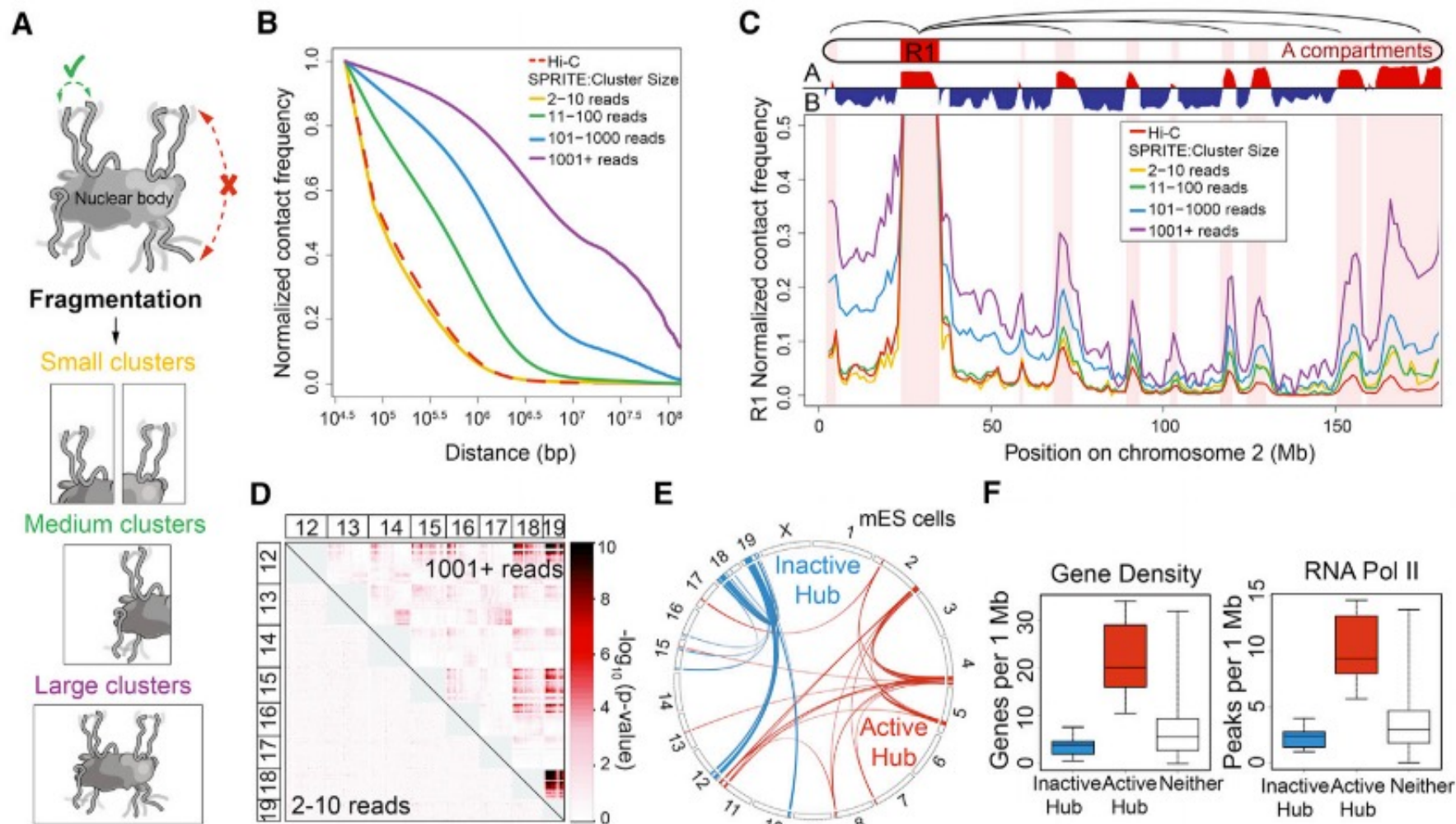
SPRITE

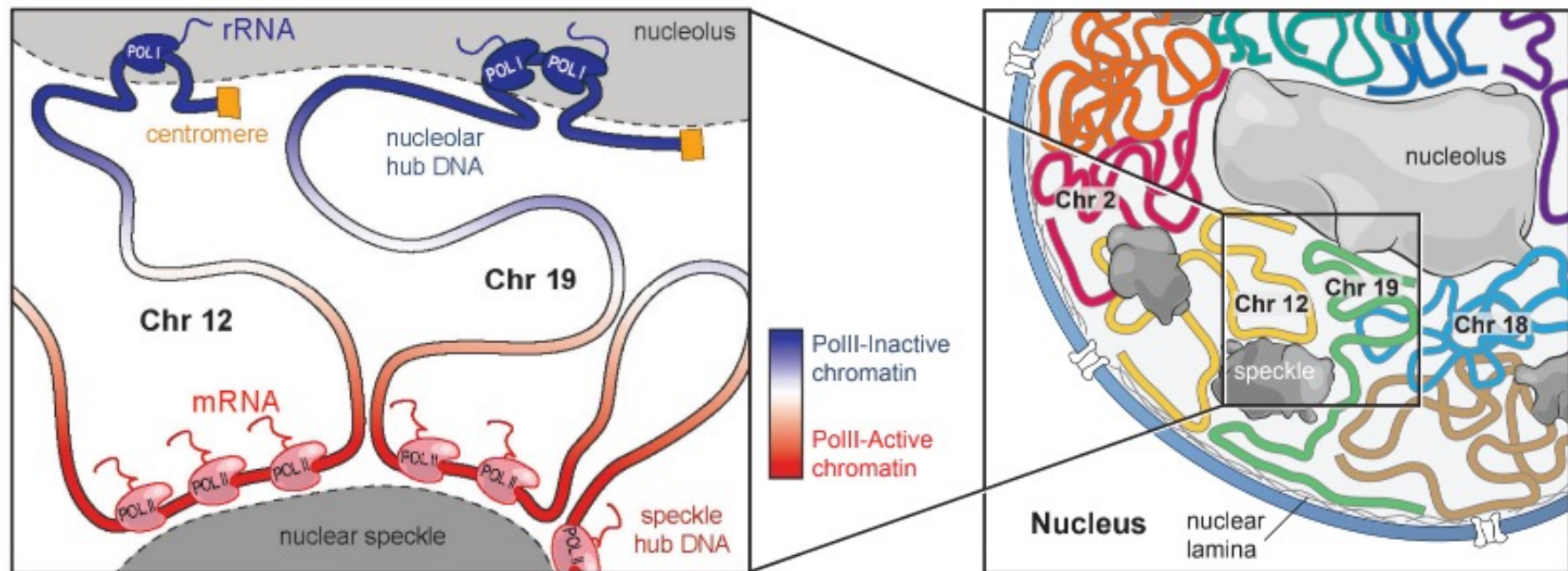
Split-Pool Recognition of Interactions by Tag Extension



A**B****C****D****GM12878 human cells****E**



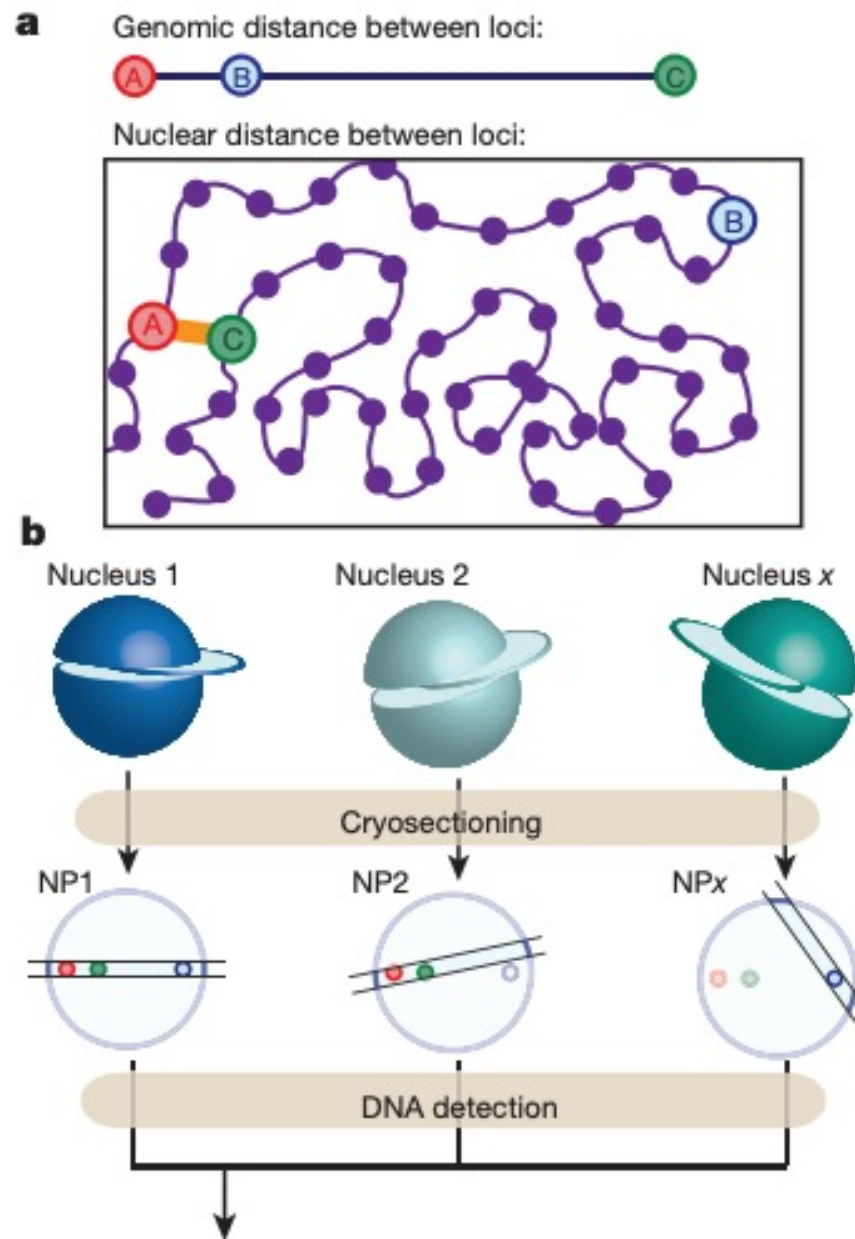




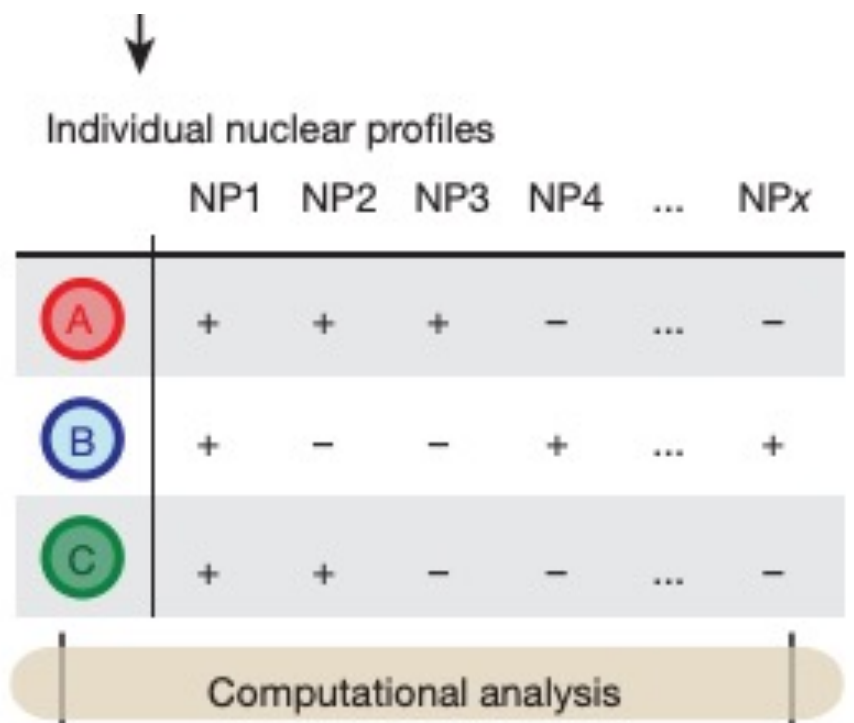
Complex multi-enhancer contacts captured by genome architecture mapping

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The organization of the genome in the nucleus and the interactions of genes with their regulatory elements are key features of transcriptional control and their disruption can cause disease. Here we report a genome-wide method, genome architecture mapping (GAM), for measuring chromatin contacts and other features of three-dimensional chromatin topology on the basis of sequencing DNA from a large collection of thin nuclear sections. We apply GAM to mouse embryonic stem cells and identify enrichment for specific interactions between active genes and enhancers across very large genomic distances using a mathematical model termed SLICE (statistical inference of co-segregation). GAM also reveals an abundance of three-way contacts across the genome, especially between regions that are highly transcribed or contain super-enhancers, providing a level of insight into genome architecture that, owing to the technical limitations of current technologies, has previously remained unattainable. Furthermore, GAM highlights a role for gene-expression-specific contacts in organizing the genome in mammalian nuclei.

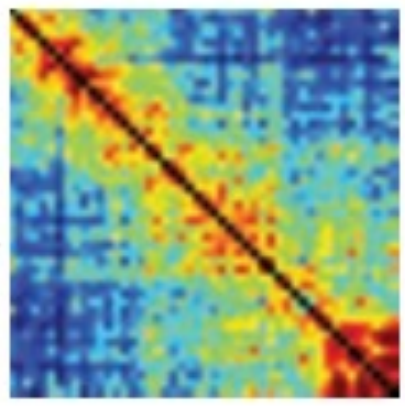


c



d

			
	-	0.01	0.09
	0.01	-	0.02
	0.09	0.02	-



Features of genome architecture

Chromatin contacts,
triplet interactions,
compaction,
radial position...

a

